

Power system-II (EE-302)

5th sem A-Section -2025

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By

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The syllabus

UNIT1: Performance of overhead transmission lines: Introduction; classification of transmission lines; performance calculation of short and medium transmission lines, Nominal T and nominal π methods; Performance Long transmission lines; Power circle diagrams (PCD).

UNIT2: Power system stability: Introduction to synchronous machine, rotor angle, Infinite bus; Definition of stability, classification of stability, power limit of transmission lines, steady state stability, transient stability, the swing equations, equal area criterion, calculation of critical clearing angles, Calculation of power angle curves for fault and post fault conditions for various types of faults, dynamic stability, factors effecting stability.

Unit3: Control of active and reactive power: Active power and frequency control: fundamental speed governing system, Governor speed regulation parameter, , Fundamentals of automatic generation control, Frequency bias, Primary and secondary control, Basic simulation models of automatic generation control; Reactive power and voltage control: Production of absorber of reactive power, methods of voltage control; shunt reactors, series reactors, synchronous condensers, static VAR system, tap changing transformers.

Unit4: Transients in power systems: Lightning phenomenon, Switching surges, travelling waves, shape and specifications of travelling waves, attenuation and distortions of travelling waves, alteration due to corona, behaviour of travelling waves at open, short and joints of overhead lines and cables, construction of lattice diagrams.

Unit5: Load flow studies: Introduction to load flow study, YBUS , formulation of load flow problems, methods.

Unit6: Economic operation of steam Power plant: Introduction; Methods of loading turbo - generators, Thermal plant cost modelling, Input - output curves, incremental cost, cost curve: Linear and quadratic, method of Lagrangion multiplier, Equality constraints and inequality constraints, transmission loss, optimum generator allocation with and without transmission loss; Penalty factors, iterative procedure to solve co-ordination equation. Advantages of combined operation with hydro plants, Introduction with a schematic diagram having cost and load curve.

Books

1. C.L. Wadhwa, Electrical Power systems, Wiley Eastern
2. Ashfaq Hussain, Electrical Power System, CBS Publishers
3. B.R.Gupta, Generation of Electrical Energy, S. Chand
4. Soni, Gupta, Bhatnagar, Electric Power, Dhanpat Rai & Sons
5. J.B.Gupta, A course in Power Systems, S.K.KATIA & SONS
6. P.Kundur, Power system stability and control, McGraw-Hill
7. O.I.Elgerd, Electric Energy System Theory, Tata Mcgraw Hilll

Course Outcomes (Cos):

At the end of the course the students will be able to

1. define various performance indices of a power system network and describe the importance of each.
2. Impart the knowledge to design transmission lines and analyse the power control methodologies in electrical networks.
3. summarize the transient phenomena, load flow studies and stability issues in power system networks.
4. explain the economic operation of thermal and hydro power plants.

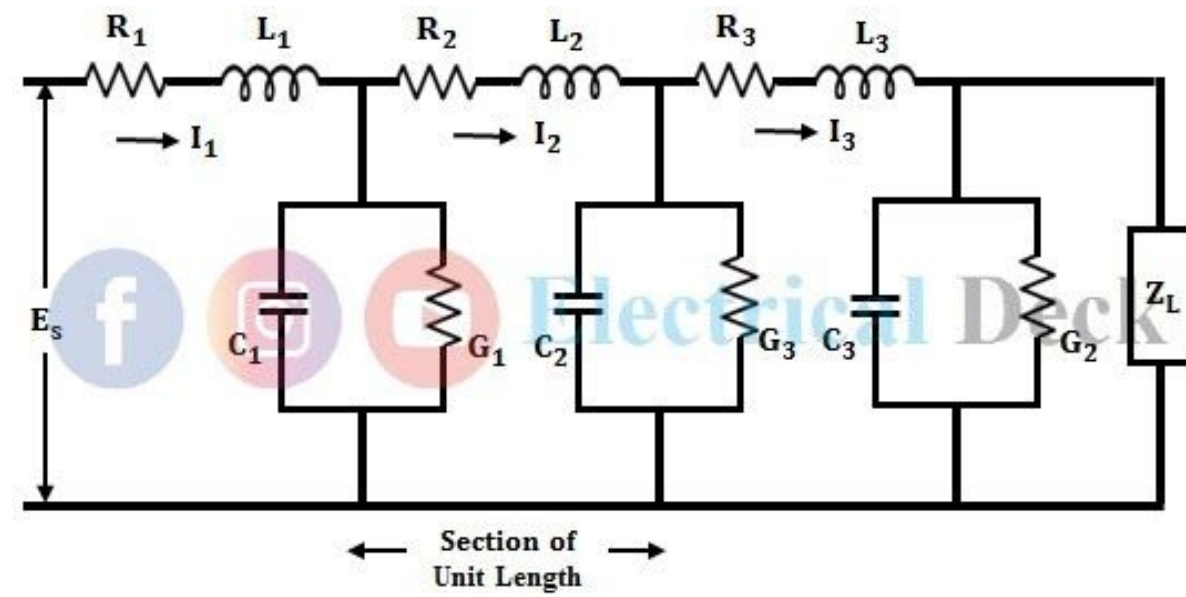
UNIT1: Performance of overhead transmission lines: Introduction; classification of transmission lines; performance calculation of short and medium transmission lines, Nominal T and nominal π methods; Performance Long transmission lines; Power circle diagrams (PCD).

Introduction:

- The term performance includes the calculations of sending end voltage, sending end current, sending end power factor, power loss in the line, transmission efficiency, voltage regulation, and the limit of power flow during steady state and transient conditions. The values of receiving end voltage, receiving end current, and receiving end power factors are generally known.
- The performance of the power system mainly depends on the performance of the transmission line in the system. A transmission line is mainly an energy corridor in the power system.
- The important considerations in the operation of transmission lines are voltage drop and power loss occurring in the line, and transmission efficiency.

- The performance of TL is governed by the parameters R , L , C , and G
- Due to R , the conductor offers opposition to the flow of current.
- L is because of the fact that a current-carrying conductor is surrounded by magnetic lines of force.
- C is because the current-carrying conductors form a capacitor with earth, between two line conductors.

- ❖ The R causes a voltage drop (IR Drop) in the line, and I^2R loss
- ❖ The L also causes a voltage drop, which is quadrature with current.
- ❖ Due to C , a current is produced called the charging current, and there is a voltage drop that is quadrature with the current.



Representation of Transmission Line Parameters in Distributed Form

II] Classification of transmission lines

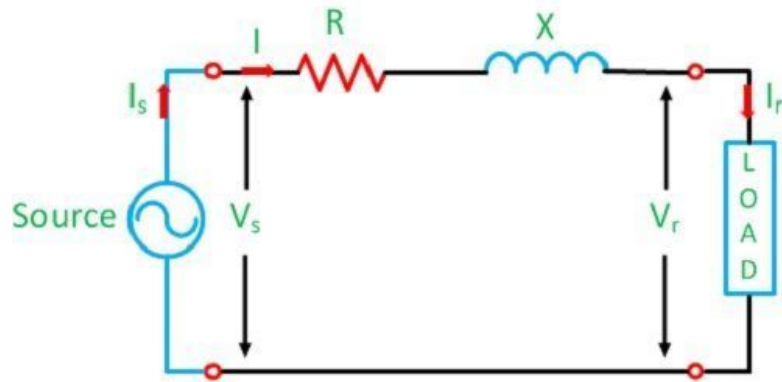
The performance of one or more of the line parameters of the line is governed by its length and conductor configurations. For example, the effect of capacitance for line lengths less than 80Km and voltage is less than 20kV, the effect of capacitance is negligible.

The transmission lines are classified into three. They are

- (i) Short,
- (ii) Medium, and
- (iii) Long transmission lines.

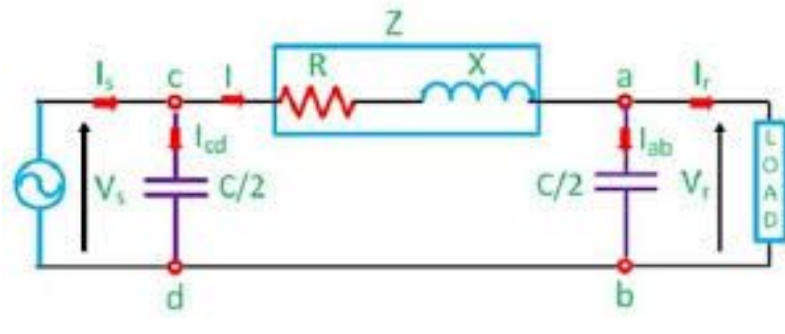
- The TLs having a length up to 80 km and an operating voltage is lower than 20 kV fall under short transmission lines.
- The TLs having a length from 80 Km to 200 (or 240) km and operating voltage between 20 kV and 100 kV is termed as medium transmission lines.
- The TLs having a length of more than 200 (or 240) km and operating above 100 kV are termed as medium transmission lines.

- The capacitance in short TL is very small and neglected. Similarly, G is also neglected in short TL. The performance of a short line mainly depends on series resistances and series inductances.
- Owing to the appreciable length and voltage of the line, the charging current is appreciable, and the capacitance effect cannot be neglected in medium lines.
- For long TLs, the parameters cannot be considered lumped because they may provide inaccurate results in performance calculations.



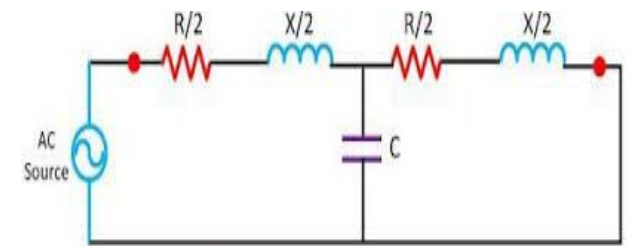
Equivalent circuit model of a short line

Circuit Globe



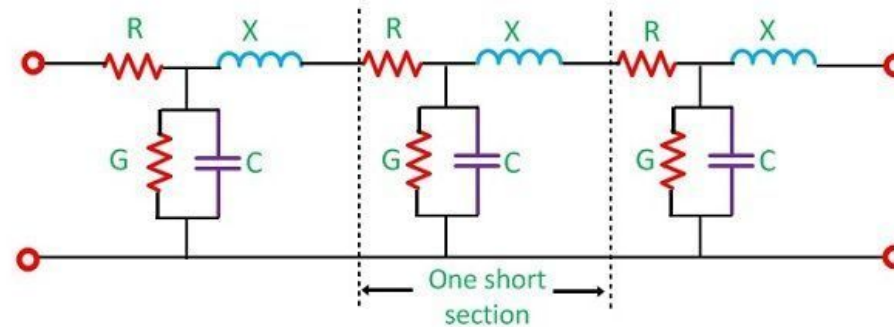
Nominal Pi model of a medium line.

Circuit Globe



Medium Transmission Line

Circuit Globe



Long transmission lines showing distributed parameters.

Circuit Globe

III] Some Performance parameters

- A. Voltage regulation or simply regulation:** Voltage regulation of a transmission line: It is defined as the change in voltage at the receiving end (RE) or load when the full load is thrown off, the sending end (SE) voltage or supply voltage and frequency remain unchanged. It is normally expressed as % of RE voltage.

$$\% \text{ Voltage regulation} = \frac{V_s - V_R}{V_R} \times 100\%$$

- B. Efficiency or transmission efficiency:** It is defined as the ratio of power delivered at the RE to the power sent from the SE. Mathematically, it is given by

$$\eta_T = \frac{P_R}{P_S} = \frac{V_R I_R \cos \phi_R}{V_S I_S \cos \phi_S} \times 100\%$$

$$\text{or } \eta_T = \frac{P_R}{P_S} = \frac{V_R I_R \cos \phi_R}{V_R I_R \cos \phi_R + \text{Line losses}} \times 100\%$$

Generalized network constants for transmission lines

For any transmission lines, the following relationships hold good

$$V_S = AV_R + BI_R$$

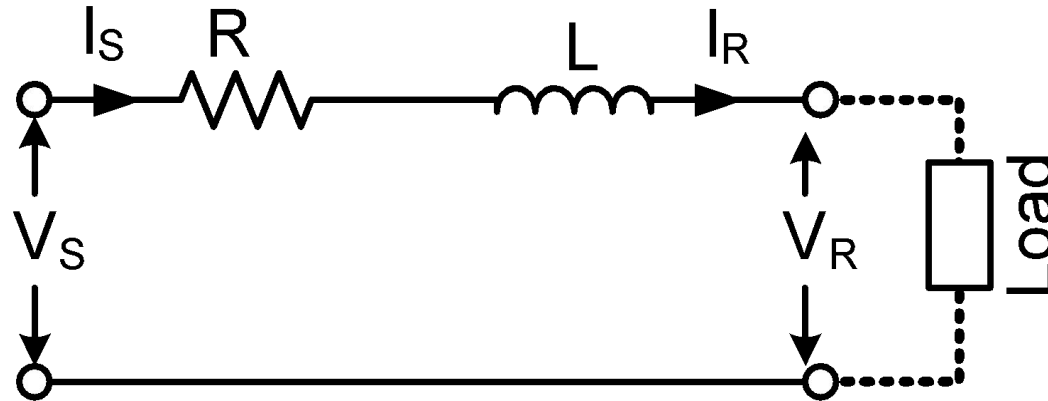
$$I_S = CV_R + DI_R$$

Where, A, B, C, D are called generalized network constants and these are related to R, X, C, G

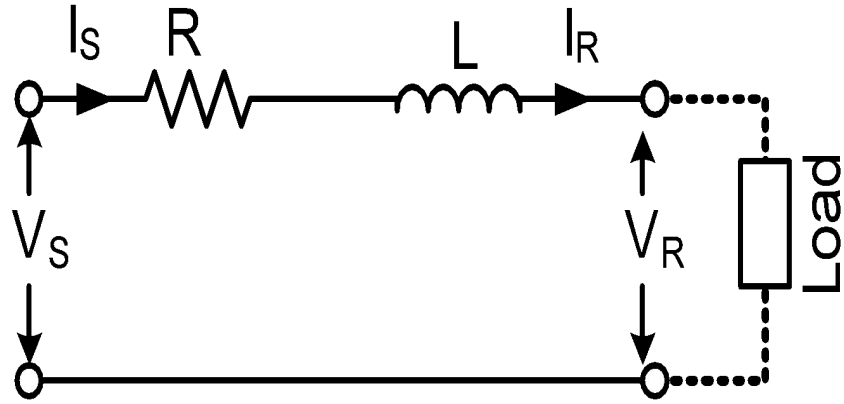
IV]. Short transmission line:

- Here, shunt conductance, G , and shunt capacitance C , are not considered because at short lengths (up to 80/100 km) and low voltage, the effects of G and C are negligible.
- Only series R and L parameters are considered.
- Though R and L are distributed uniformly along the line, in the performance calculation of a short line, they are considered lumped.
- In the lumped parameters assumption, there is no considerable error in the performance calculation of short lines.

The equivalent circuit is shown in Figure below



Analysis of Short TL



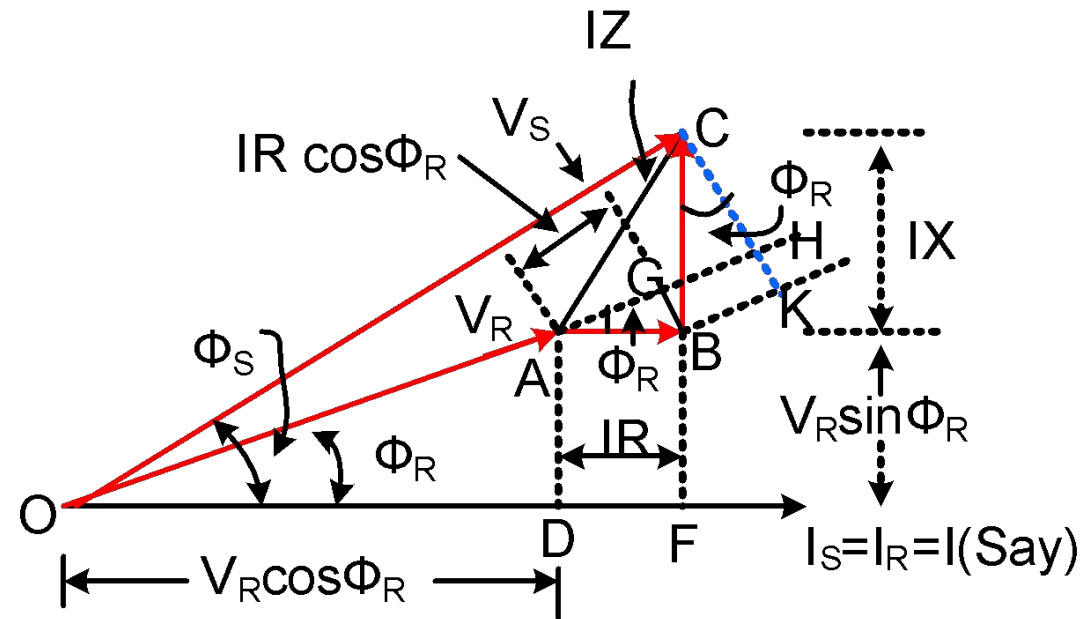
Let SE Voltage = V_s ; RE Voltage = V_R ;

SE Current = I_s ; RE Current = I_R ;

RE Power factor = $\cos\phi_R$ (lagging)

SE power factor = $\cos\phi_s$

The phasor diagram is shown in Figure



From the phasor diagram

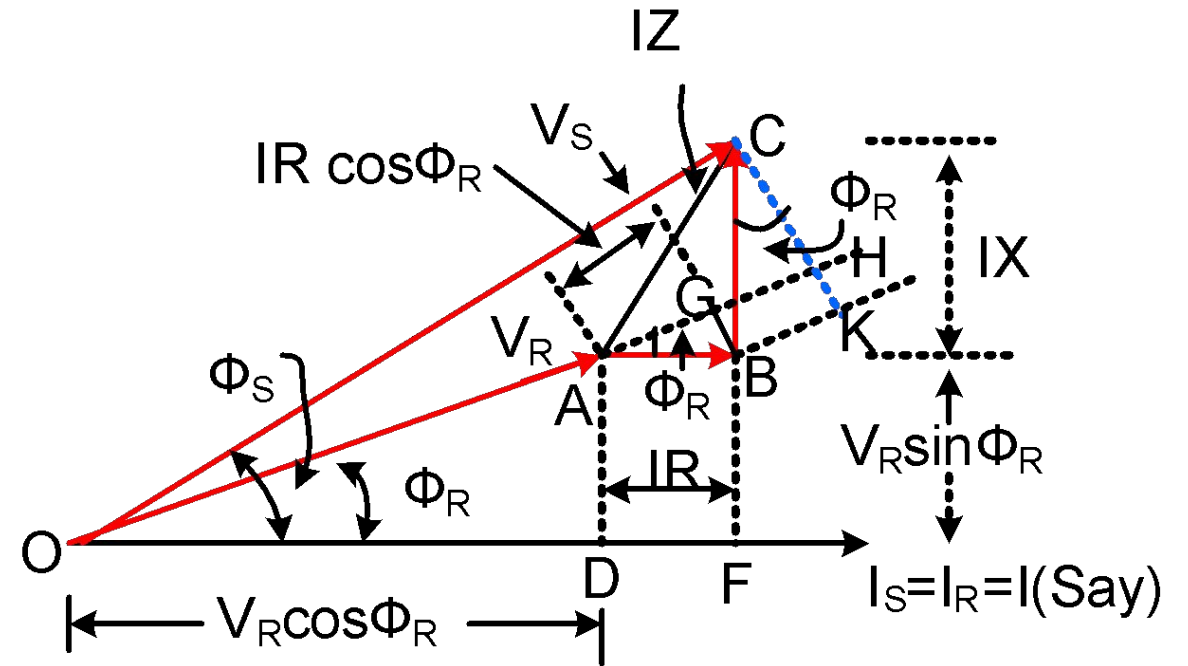
$$V_S = V_R + IR + jIX = V_R + I(R + jX) = IZ$$

$$\begin{aligned} V_S &= OC = \sqrt{OF^2 + FC^2} \\ &= \sqrt{(OD + DF)^2 + (FB + BC)^2} \\ &= \sqrt{(V_R \cos \phi_R + IR)^2 + (V_R \sin \phi_R + IX)^2} \end{aligned}$$

$$\phi_S = \tan^{-1} \left(\frac{FC}{OF} \right) = \frac{V_R \sin \phi_R + IX}{V_R \cos \phi_R + IR}$$

$$\text{SE power factor, } \cos \phi_s = \frac{OF}{OC} = \frac{V_R \cos \phi_R + IR}{V_S}$$

$$\begin{aligned} \% \text{ Voltage regulation} &= \frac{V_S - V_R}{V_R} \times 100\% \\ &= \frac{\sqrt{(V_R \cos \phi_R + IR)^2 + (V_R \sin \phi_R + IX)^2} - V_R}{V_R} \times 100\% \end{aligned}$$



$$\text{Power delivered} = V_R I_R \cos \phi_R$$

$$\text{Line losses} = I^2 R$$

$$\text{Power sent out} = V_R I_R \cos \phi_R + I^2 R$$

$$\% \text{ Transmission efficiency, } \eta_T = \frac{V_R I_R \cos \phi_R}{V_R I_R \cos \phi_R + I^2 R} \times 100\%$$

The vector diagram is drawn taking I_r , the receiving end current, as the reference.

From the vector diagram,

$$V_s \cos \phi_s = V_r \cos \phi_r + I_r R \quad (4.2a)$$

$$V_s \sin \phi_s = V_r \sin \phi_r + I_r X \quad (4.2b)$$

Squaring and adding equations (4.2a) and (4.2b),

$$V_s^2 = V_r^2 + 2I_r R V_r \cos \phi_r + 2I_r X V_r \sin \phi_r + I_r^2 (R^2 + X^2) \quad (4.3)$$

$$V_s = V_r \sqrt{1 + \frac{2I_r R \cos \phi_r}{V_r} + \frac{2I_r X \sin \phi_r}{V_r} + \frac{I_r^2}{V_r^2} (R^2 + X^2)}$$

In practice the last term under the square root sign is generally negligible; therefore,

$$V_s = V_r \left\{ 1 + \left(\frac{2I_r R}{V_r} \cos \phi_r + \frac{2I_r X}{V_r} \sin \phi_r \right) \right\}^{1/2} \quad (4.4)$$

The terms within the simple brackets is small as compared to unity. Using binomial expansion and limiting only to second term,

$$V_s \simeq V_r + I_r R \cos \phi_r + I_r X \sin \phi_r \quad (4.5)$$

Here V_s is the sending end voltage corresponding to a particular load current and power factor condition. It can be seen from the equivalent circuit of short line that the receiving end voltage under no load V_r' is the same as the sending end voltage under full load condition, *i.e.*,

$$V_r' = V_s$$

$$\begin{aligned} \therefore \quad \% \text{ regulation} &= \frac{V_s - V_r}{V_r} \times 100 \\ &= \left(\frac{I_r R}{V_r} \cos \phi_r + \frac{I_r X}{V_r} \sin \phi_r \right) \times 100 \end{aligned} \quad (4.6)$$

$$\begin{aligned} \text{or} \quad \text{regulation per unit} &= \frac{I_r R}{V_r} \cos \phi_r + \frac{I_r X}{V_r} \sin \phi_r \\ &= v_r \cos \phi_r + v_x \sin \phi_r \end{aligned} \quad (4.7)$$

where v_r and v_x are the per unit values of resistance and reactance of the line.

$$V_s = AV_r + BI_r \quad (4.8)$$

$$I_s = CV_r + DI_r \quad (4.9)$$

where A , B , C , D are called the constants of the network. The transmission line is also a four-terminal network and it is now desired to find these constants for short transmission line.

Before these constants are determined it is desirable to understand what these constants are.

From equation (4.8),

$$A = \left. \frac{V_s}{V_r} \right|_{I_r = 0}$$

This means A is the voltage impressed at the sending end per volt at the receiving end when receiving end is open. It is dimensionless.

$$B = \left. \frac{V_s}{I_r} \right|_{V_r = 0}$$

B is the voltage impressed at the sending end to have one ampere at the short circuited receiving end. This is known as transfer impedance in network theory.

From equation (4.9),

$$C = \left. \frac{I_s}{V_r} \right|_{I_r = 0}$$

C is the current in amperes into the sending end per volt on the open-circuited receiving end. It has the dimension of admittance.

$$D = \left. \frac{I_s}{I_r} \right|_{V_r = 0}$$

D is the current at the sending end for one ampere of current at the short circuited receiving end. The constants A , B , C and D are related for a passive network as follows:

$$AD - BC = 1$$

This relation provides a good check on the values of these constants.

The sending end voltage and current can be written from the equivalent network as

$$V_s = V_r + I_r Z \quad (4.10)$$

$$I_s = I_r \quad (4.11)$$

Comparing the coefficients of equations (4.10) and (4.11) with equations (4.8) and (4.9) respectively, the constants for short transmission line are

$$A = 1$$

$$B = Z$$

$$C = 0$$

$$D = 1$$

Checking the values of A , B , C and D with the relation

$$AD - BC = 1$$

$$1 \cdot 1 - Z \cdot 0 = 1$$

So, the values calculated are correct for a short transmission line.

The $ABCD$ constants can be used for calculation of regulation of the line as follows:

Normally the quantities P , I_r and $\cos \phi_r$ at the receiving end are given and of course the $ABCD$ constant. Then determine sending end voltage using relation

$$V_s = AV_r + BI_r$$

To determine V_r' the no load voltage at the receiving end, equation (4.8) is made use of

$$V_r' = \frac{V_s}{A}, \text{ when } I_r = 0$$

$$\% \text{ regulation} = \frac{V_s / A - V_r}{V_r} \times 100$$

is thus evaluated.

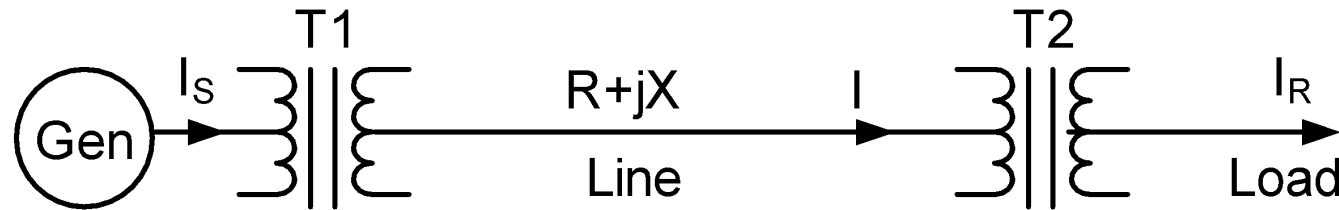
To determine $\% \eta$ of transmission, the following relation is made use of:

$$\begin{aligned} \% \eta &= \frac{\text{Power received at the receiving end}}{\text{Power received at the receiving end} + \text{losses}} \times 100 \\ &= \frac{P}{P + 3I_r^2 R} \times 100 \end{aligned}$$

where R is the resistance per phase of the line.

Line with transformer

Transformers are installed at both ends of a transmission line. Step up transformer at the SE and step down transformer at the RE. Fig shows such a line.



Let r_1 and r_2 be the resistances of the high and low voltage windings to neutral, x_1 and x_2 be the leakage reactance in the high and low voltage windings to neutral, N_1 and N_2 be the numbers of turns in the high and low voltage windings, respectively.

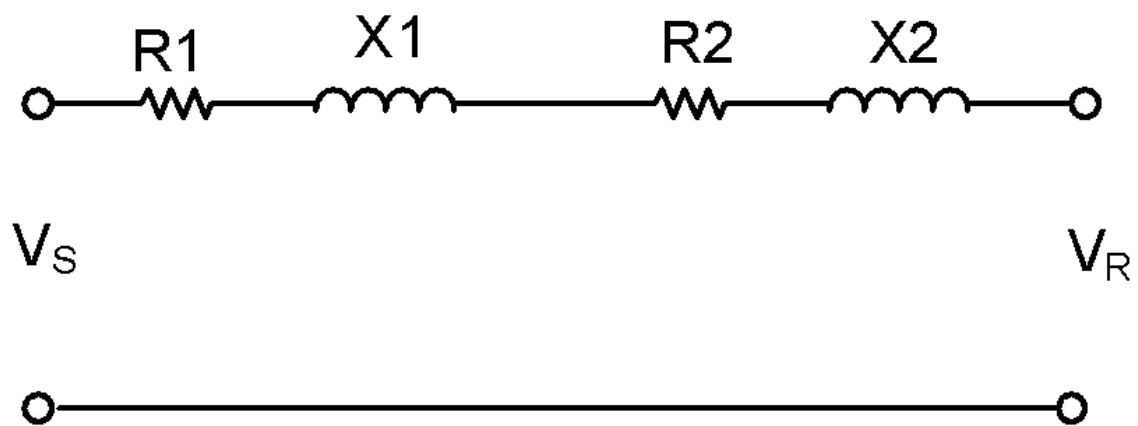
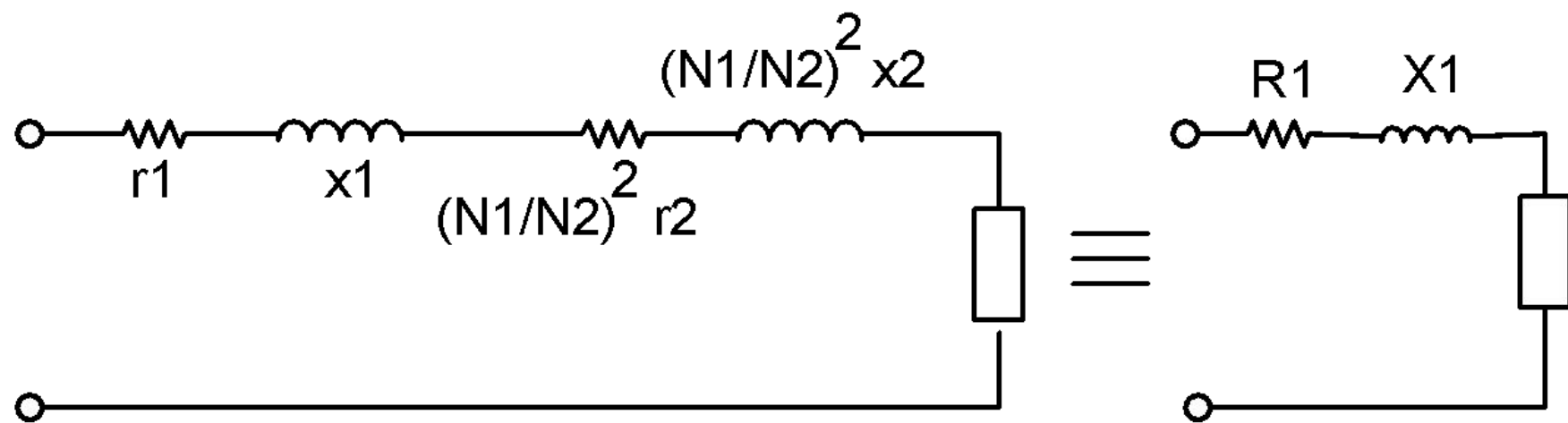
The equivalent resistance referred to high voltage side, $R_1 = r_1 + \left(\frac{N_1}{N_2}\right)^2 r_2$

The equivalent reactance referred to high voltage side, $X_1 = x_1 + \left(\frac{N_1}{N_2}\right)^2 x_2$

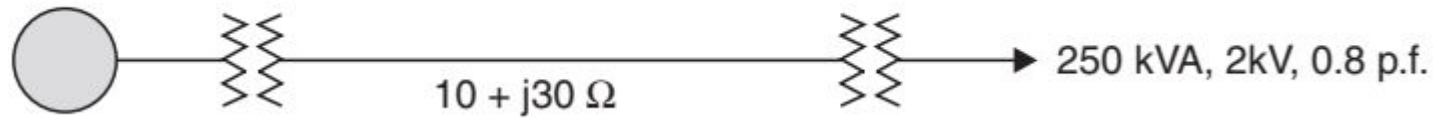
The total resistance of the line and the transformer, $R_T = R_1 + R + R_2$

The total reactance of the line and the transformer, $X_T = X_1 + X + X_2$

Here, R_2 and X_2 are the equivalent resistance referred to the high-voltage side of the load-end transformer.



Example 4.1: Determine the voltage at the generating station and the efficiency of transmission for the following 1-phase system



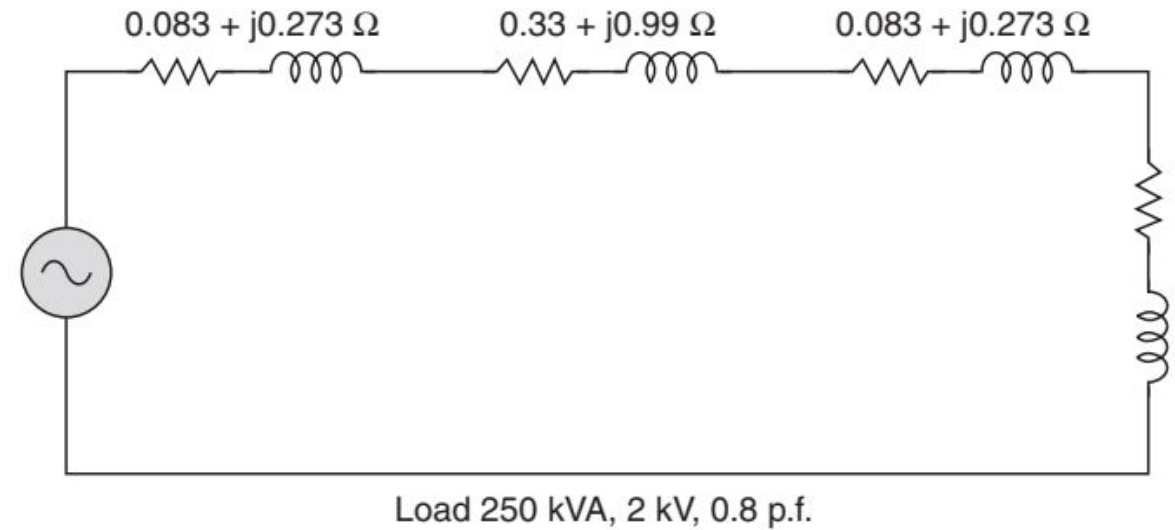
Transformer ratio 2 kV/11 kV. The resistance on l.v. side = 0.04 ohm and h.v. side 1.3 ohm. Reactance on l.v. and h.v. side is 0.125 ohm and 4.5 ohm.

Solution: The transmission line equivalent impedance when referred to l.v. side will be

$$Z = 10 \times \left(\frac{2}{11}\right)^2 + j30 \times \left(\frac{2}{11}\right)^2$$
$$= (0.33 + j0.99)$$

$$\text{Transformer impedance} = 0.04 + 1.3 \times \left(\frac{2}{11}\right)^2 + j0.125 + j4.5 \times \left(\frac{2}{11}\right)^2$$
$$= (0.083 + j0.273)$$

The equivalent circuit for the total system



The line current $= \frac{250 \times 1000}{2000} = 125$ amps.

The line loss $= I^2 R = 125^2 \times 0.496 = 7.7$ kW

The output $= 250 \times 0.8 = 200$ kW

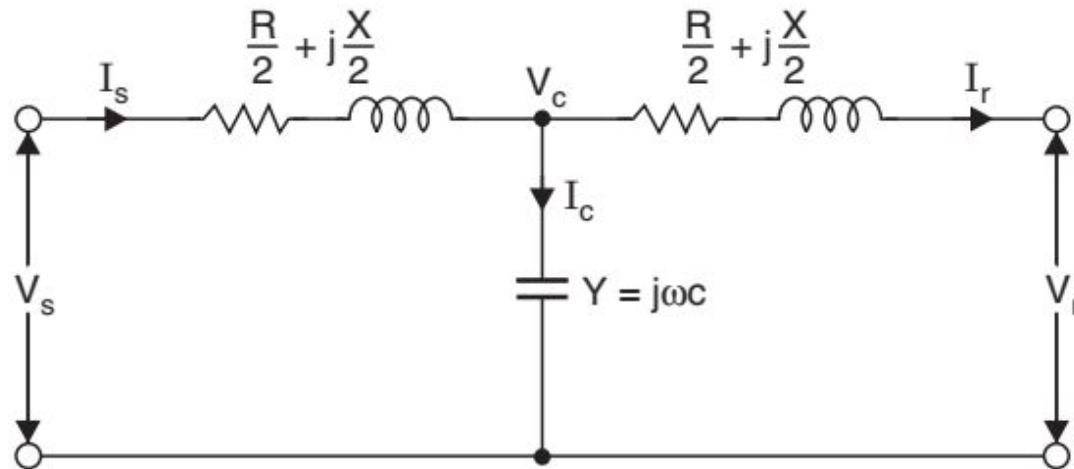
$\therefore \% \eta = \frac{200}{200 + 7.7} \times 100 = 96.3\%.$ **Ans.**

Taking I_r as the reference, the sending end voltage

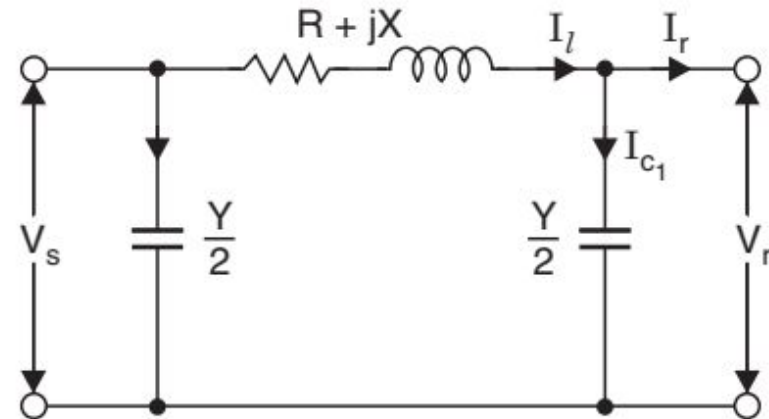
$$\begin{aligned} V_s &= (V_r \cos \phi_r + IR) + j(V_r \sin \phi_r + IX) \\ &= (2000 \times 0.8 + 125 \times 0.496) + j(2000 \times 0.6 + 125 \times 1.536) \\ &= 2168 \text{ volts. } \mathbf{Ans.} \end{aligned}$$

V]. Medium lines

- The TLs having a length from 80 Km to 200 (or 240) km and operating voltage between 20 kV and 100 kV is termed as medium transmission lines.
- Owing to the appreciable length and voltage of the line, the charging current is appreciable, and the capacitance effect cannot be neglected in medium lines.
- In medium-length lines, where the parameters are assumed to be lumped.
- The shunt capacitance is either assumed to be concentrated at the middle of the line or half of the total capacitance is concentrated at each end of the line.
- The two configurations are known as nominal-T and nominal- π , respectively. The nominal circuits are shown in the figure.



(a) Nominal-T

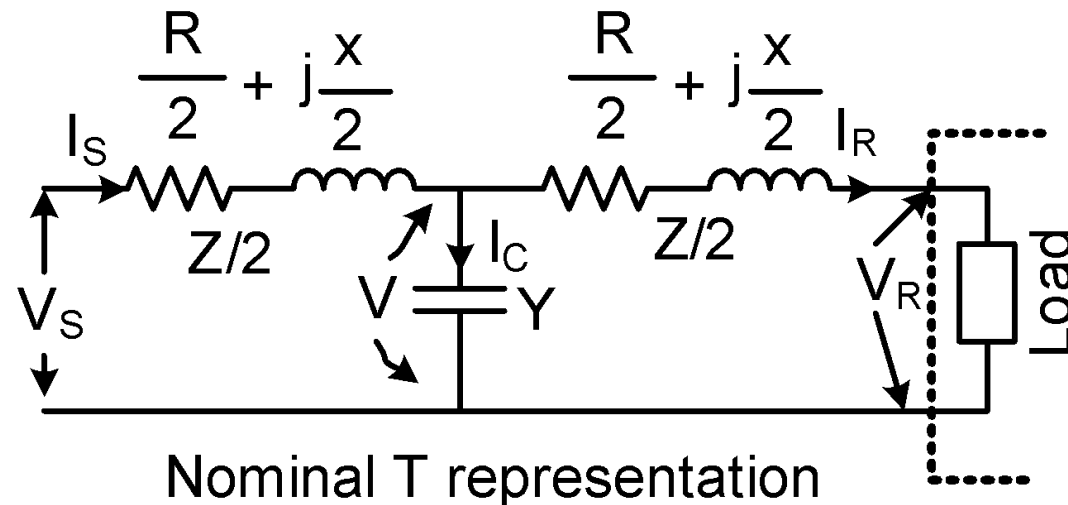


(b) Nominal- π

- It is to be noted that the two representations are approximate to the exact representation of the actual line.
- Also, the two representations are not equivalent, as can be seen by using the star-delta transformations.
- However, they are good enough for practical purposes and do not involve much error.
- They are only for academic interests. But when long lines are concerned, there are considerable errors because actual distributed parameters are assumed lumped.

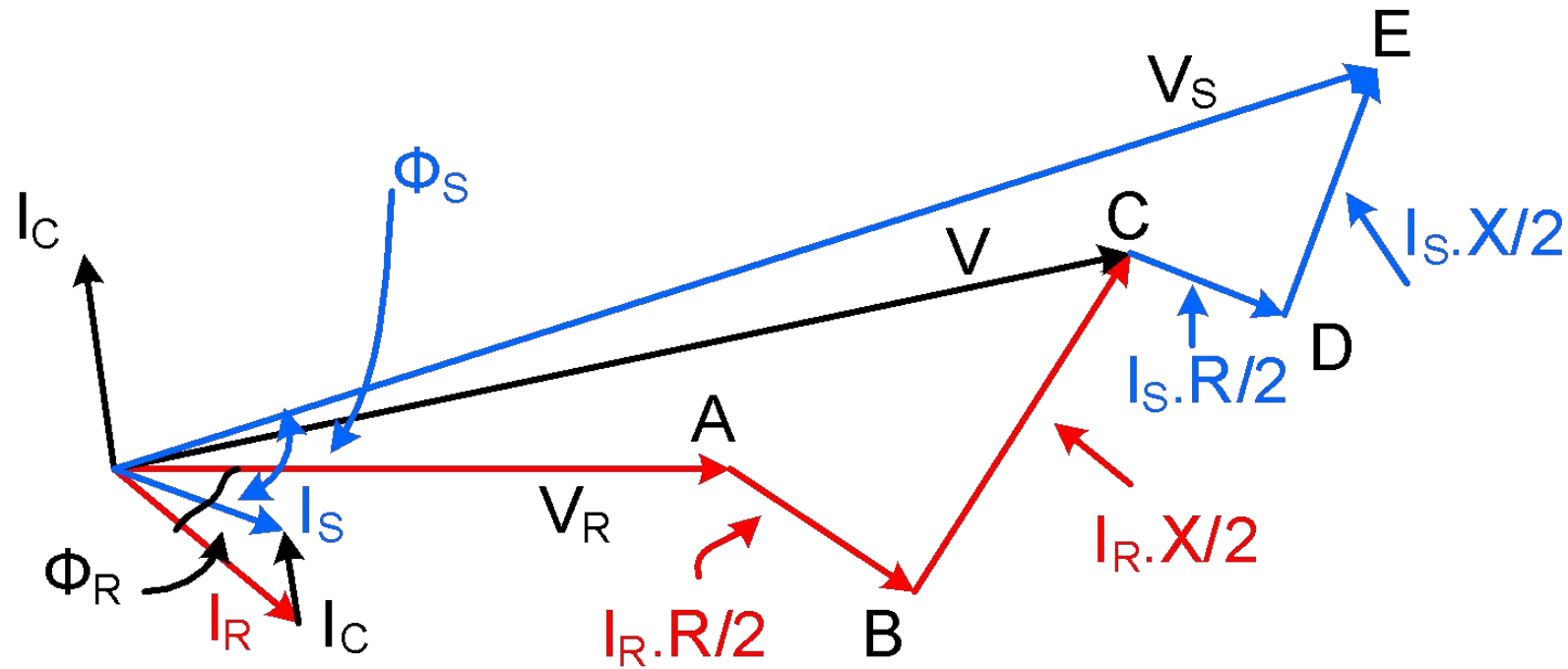
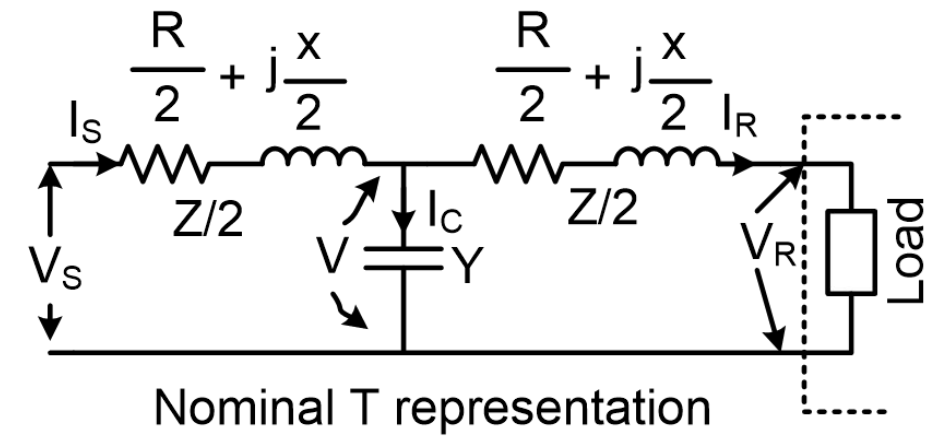
Nominal T representation of medium TLs

- The capacitance of each conductor is represented by an equivalent capacitance shunted between the conductor and the neutral midway along the line.
- The total resistance, R and Total reactance, X , both are equally divided into two, one of which is placed at the left side arm of a T and other half is placed at the right half of the T. The shunt capacitance, C is placed along the vertical arm of a the T.
- The equivalent circuit is shown in the Figure.



Analysis of Nominal T representation

Phasor diagram



Denoting line impedance, $Z=R+jX$, and hence $Z/2=R/2+jX/2$

$$\bar{V} = \bar{V}_R + \bar{I}_R \left(\frac{\bar{Z}}{2} \right)$$

$$\bar{I}_C = \bar{Y}\bar{V} = \bar{Y} \left(\bar{V}_R + \bar{I}_R \left(\frac{\bar{Z}}{2} \right) \right)$$

And so, $\bar{I}_S = \bar{I}_R + \bar{I}_C$

$$= \bar{I}_R + \bar{Y}\bar{V}_R + \bar{I}_R \frac{\bar{Y}\bar{Z}}{2}$$

$$= \bar{I}_R \left(1 + \frac{\bar{Y}\bar{Z}}{2} \right) + \bar{Y}\bar{V}_R$$

Or simply, $I_S = I_R \left(1 + \frac{YZ}{2} \right) + YV_R$

$$\text{Also, } V_S = V + I_S \cdot \frac{Z}{2} = V_R + I_R \cdot \frac{Z}{2} + \left(I_R \left(1 + \frac{YZ}{2} \right) + YV_R \right) \cdot \frac{Z}{2}$$

$$= V_R \left(1 + \frac{YZ}{2} \right) + I_R \left(\frac{Z}{2} + \frac{Z}{2} + \frac{YZ^2}{4} \right)$$

$$= V_R \left(1 + \frac{YZ}{2} \right) + I_R Z \left(1 + \frac{YZ}{4} \right)$$

We can now find that

$$A = D = 1 + \frac{YZ}{2}; B = Z \left(1 + \frac{YZ}{4} \right); C = Y$$

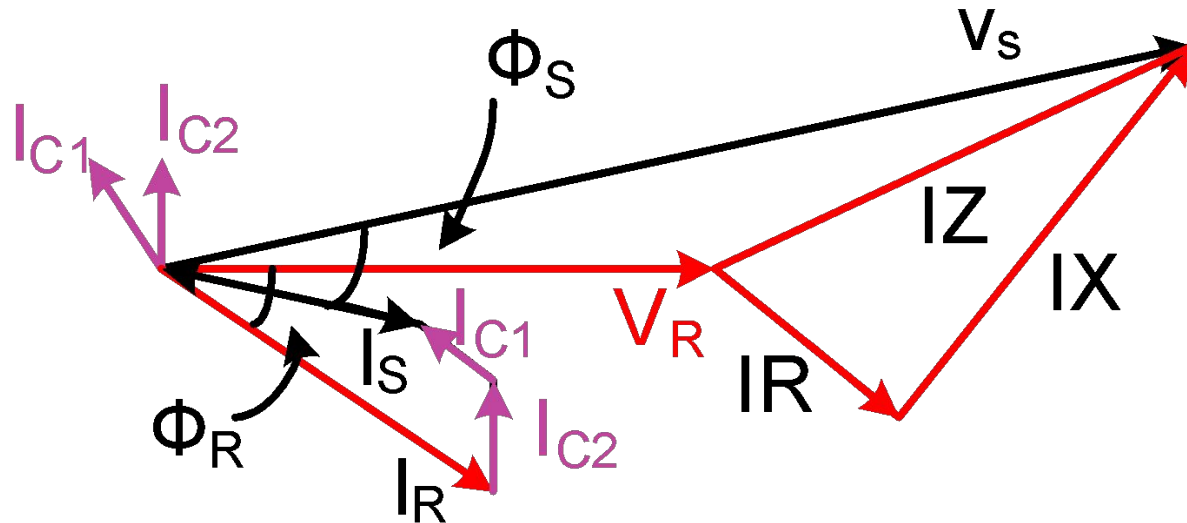
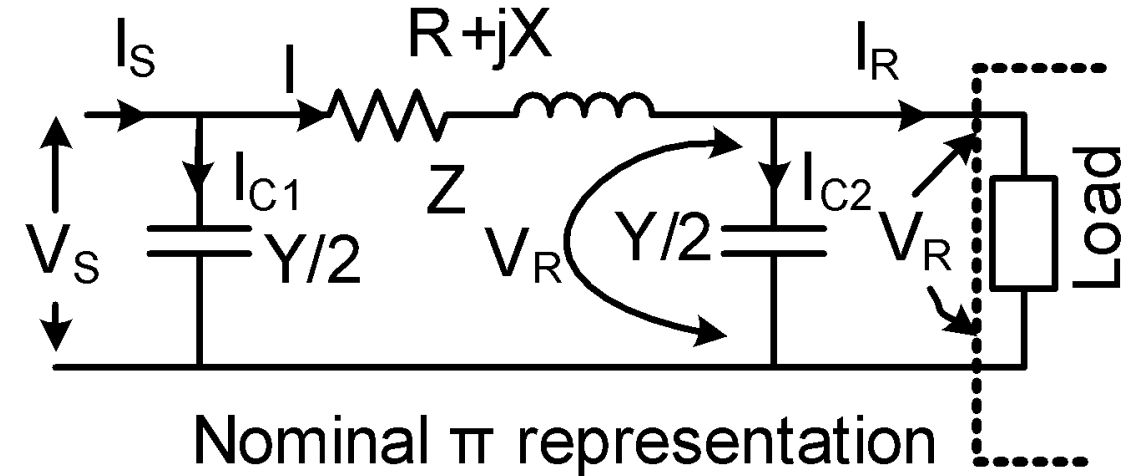
$$\text{and } AD - BC = \left(1 + \frac{YZ}{2} \right)^2 - YZ \left(1 + \frac{YZ}{4} \right) = 1$$

Rearranging the terms

$$\begin{bmatrix} V_S \\ I_S \end{bmatrix} = \begin{bmatrix} 1 + \frac{YZ}{2} & Z \left(1 + \frac{YZ}{4} \right) \\ Y & 1 + \frac{YZ}{2} \end{bmatrix} \begin{bmatrix} V_R \\ I_R \end{bmatrix}$$

Nominal π representation of TLs

- Here one half of the total capacitance of each conductor is shunted at the end between the conductor and neutral at each end of the line.
- The equivalent circuit is shown in the Figure.



Here,

$$I_{C2} = \frac{Y}{2} \cdot V_R; I = I_R + I_{C2} = I_R + \frac{Y}{2} \cdot V_R$$

$$\text{So that, } V_S = V_R + ZI = V_R + Z \left(I_R + \frac{Y}{2} \cdot V_R \right)$$

$$\text{Or } V_S = V_R \left(1 + \frac{YZ}{2} \right) + I_R Z$$

$$\text{Again, } I_{C1} = \frac{Y}{2} \cdot V_S = \frac{Y}{2} \left(V_R \left(1 + \frac{YZ}{2} \right) + I_R Z \right)$$

$$I_S = I + I_{C1} = I_R + \frac{Y}{2} V_R + \frac{Y}{2} \left(V_R \left(1 + \frac{YZ}{2} \right) + I_R Z \right)$$

$$\text{Or } I_S = V_R Y \left(1 + \frac{YZ}{2} \right) + I_R \left(1 + \frac{YZ}{2} \right)$$

Comparing with the general network equations

$$V_S = AV_R + BI_R$$

$$I_S = CV_R + DI_R$$

$$A = D = 1 + \frac{YZ}{2}$$

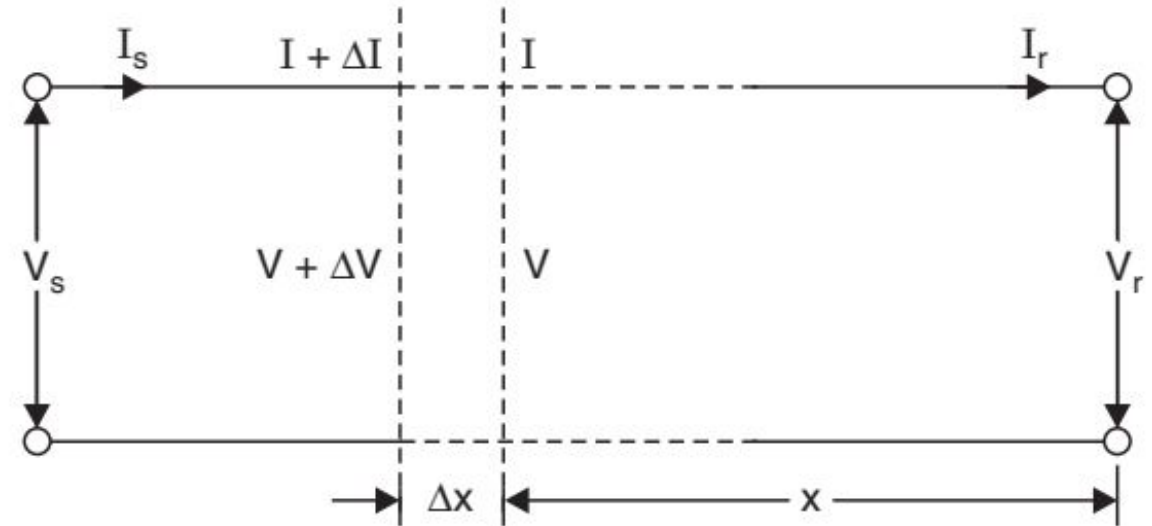
$$B = Z; C = Y \left(1 + \frac{YZ}{4} \right)$$

$$\therefore \begin{pmatrix} V_S \\ I_S \end{pmatrix} = \begin{pmatrix} 1 + \frac{YZ}{2} & Z \\ Y \left(1 + \frac{YZ}{4} \right) & 1 + \frac{YZ}{2} \end{pmatrix} \begin{pmatrix} V_R \\ I_R \end{pmatrix}$$

Long transmission line

- So far electrically short transmission lines less than 160 km in length have been considered wherein the parameters are assumed to be lumped.
- In case the lines are more than 160 km/200km long, for accurate solutions the parameters must be taken as distributed uniformly along the length as a result of which the voltages and currents will vary from point to point on the line.
- Consider the figure for analysis.

Let z = series impedance per unit length
 y = shunt admittance per unit length
 l = length of the line
 $Z = zl$ = total series impedance
 $Y = yl$ = total shunt admittance



For clarity the elemental length dx is redrawn

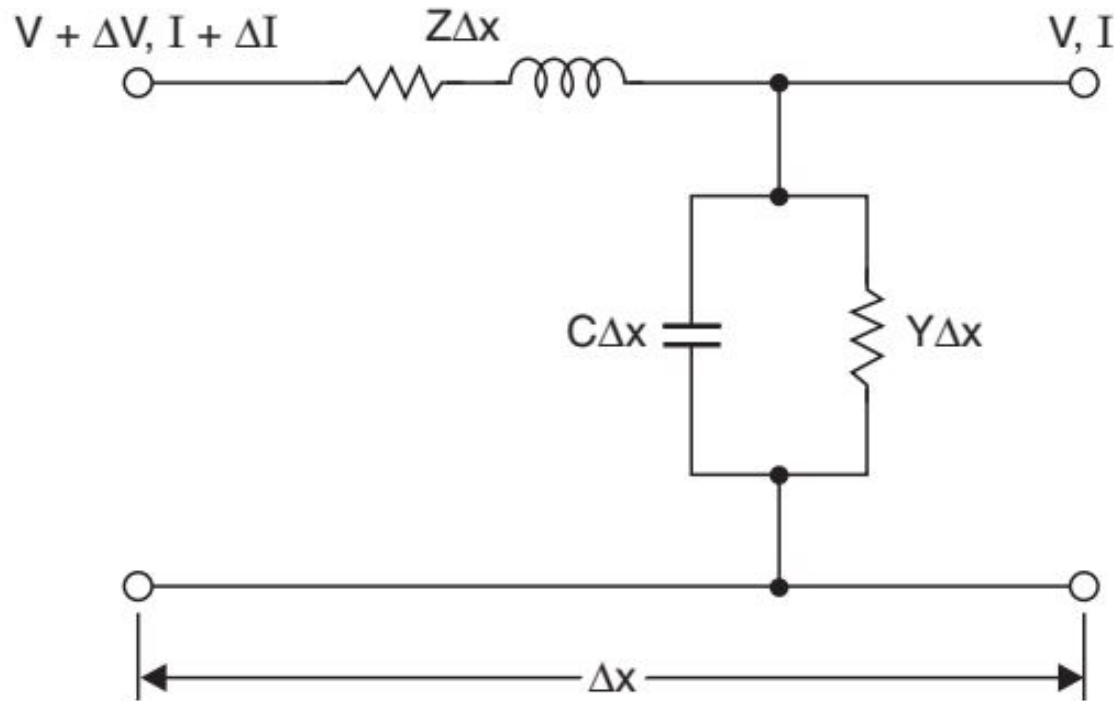


Fig. 4.11 Equivalent of a differential length of a line.

- For analysis we shall take the receiving end as the reference for measuring distances.
- Take an elemental length dx of the line at a distance of x from the receiving end.
- Say the voltage and current at a distance x are V and I and at a distance $x + dx$, $V + \Delta V$ and $I + \Delta I$ respectively.

$$\begin{aligned}\Delta V &= Iz \Delta x \\ \Delta I &= Vy \Delta x\end{aligned}\quad (4.26)$$

From equation (4.26)

$$\begin{aligned}\frac{\Delta V}{\Delta x} &= Iz \\ \frac{\Delta I}{\Delta x} &= Vy\end{aligned}$$

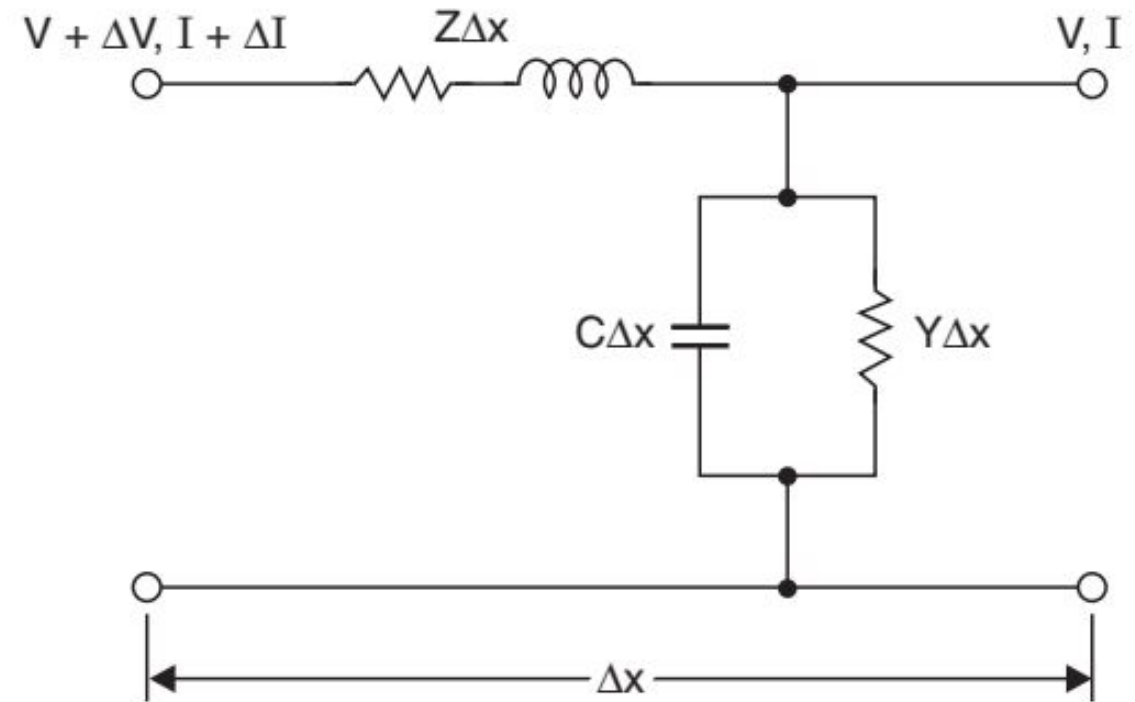


Fig. 4.11 Equivalent of a differential length of a line.

which in the limit when $\Delta x \rightarrow 0$ reduce to

$$\frac{dV}{dx} = Iz \quad (4.28)$$

$$\frac{dI}{dx} = Vy \quad (4.29)$$

Differentiating equation (4.28), we get

$$\frac{d^2V}{dx^2} = z \frac{dI}{dx} = z \cdot y \cdot V$$

or

$$\frac{d^2V}{dx^2} - zyV = 0 \quad (4.30)$$

The solution of this equation is

$$V = A \exp (\sqrt{yz} \cdot x) + B \exp (-\sqrt{yz} \cdot x) \quad (4.31)$$

A and B are arbitrary constants whose values depend on boundary conditions, that is A and B are found from the known values of V and I at some points of the line.

Differentiating the equation of voltage with respect to x

$$V = Ae^{\gamma x} + Be^{-\gamma x}$$

$$\frac{dV}{dx} = A\gamma e^{\gamma x} - B\gamma e^{-\gamma x}$$

$$\text{Thus, } \frac{dV}{dx} = A\gamma e^{\gamma x} - B\gamma e^{-\gamma x} = Iz$$

$$\text{from which we will get, } I = \frac{\sqrt{zy}}{z} (A\gamma e^{\gamma x} - B\gamma e^{-\gamma x}) = \sqrt{\frac{y}{z}} (A\gamma e^{\gamma x} - B\gamma e^{-\gamma x})$$

Now, from equations (4.28) and (4.31) let

$$Z_c = \sqrt{\frac{z}{y}} \text{ and } \gamma = \sqrt{yz} = \alpha + j\beta \quad (4.32)$$

where Z_c is known as characteristic impedance and γ the propagation constant.

The equations (4.31) and (4.32) are rewritten as

$$V = Ae^{\gamma x} + Be^{-\gamma x} \quad (4.33)$$

$$I = \frac{I}{Z_c} (Ae^{\gamma x} - Be^{-\gamma x}) \quad (4.34)$$

Two constants are to be determined, hence two boundary conditions should be known. As mentioned previously the receiving end voltage and current are known.

$\therefore$ At $x = 0$,

$$V = V_r \quad \text{and} \quad I = I_r$$

Substituting these values in equations (4.33) and (4.34),

$$V_r = A + B$$

$$I_r = \frac{1}{Z_c} (A - B)$$

$$A = \frac{V_r + I_r Z_c}{2} \text{ and } B = \frac{V_r - I_r Z_c}{2}$$

$\mathbb{Z}$

$$V = \frac{V_r + I_r Z_c}{2} e^{\gamma x} + \frac{V_r - I_r Z_c}{2} e^{-\gamma x} \quad (4.35)$$

and

$$I = \frac{1}{Z_c} \left[\frac{V_r + I_r Z_c}{2} e^{\gamma x} - \frac{V_r - I_r Z_c}{2} e^{-\gamma x} \right] \quad (4.36)$$

As mentioned previously V and I are the voltage and current at any point distant x from the receiving end.

It can be seen very easily from the above expression that V and I (magnitude and phase) are functions of the distance x , receiving end voltage V_r and current I_r and the parameters of the line, which means they vary as we move from receiving end towards the sending end.

Before we proceed further to determine the equivalent circuit for a long transmission line it looks imperative to understand the physical significance of the voltage and current equations (4.35) and (4.36).

The quantities Z_c and γ are complex.

$$Z_c = \sqrt{\frac{z}{y}} = \sqrt{\frac{r + j\omega L}{g + j\omega C}}$$

For a lossless line $r = 0$, $g = 0$,

$$Z_c = \sqrt{\frac{L}{C}}$$

a pure resistance, and this is known as surge impedance of the line.

- When dealing with high frequencies or surges normally the losses are neglected and, therefore, the characteristic impedance becomes the surge impedance.
- Surge impedance loading of a line is the power transmitted when the line is terminated through a resistance equal to surge impedance.
- The approximate value of surge impedance for overhead lines is 400 ohms and that for cables is about 40 ohms.
- The phase angle of Z_c for transmission lines is usually between 0° and -15° . A line terminated in its characteristic impedance is called a flat line or an infinite line.
- The latter term arises from the fact that a line of infinite length cannot have a reflected wave.

- The lower value of surge impedance in case of cables is due to the relatively large capacitance and low inductance of the cables.
- The propagation constant $\gamma = \alpha + j\beta$; the real part is known as attenuation constant and the quadrature component β the phase constant and is measured in radians per unit length.

The equation (4.35) becomes

$$V = \frac{V_r + I_r Z_c}{2} e^{\alpha x} \cdot e^{j\beta x} + \frac{V_r - I_r Z_c}{2} e^{-\alpha x} \cdot e^{-j\beta x} \quad (4.37)$$

- The first term in the above expression is called the incident voltage wave as its value increases as x is increased.
- Since we are taking receiving end as the reference and as x increases the value of voltage increases that means a voltage wave decreases in magnitude as it travels from the sending end towards the receiving end, that is why this part of the voltage in the above expression is called incident voltage.
- For similar reason the second part is called the reflected voltage.
- At any point along the line, voltage is the sum of these two components i.e., sums of incident and reflected voltages.

As the current expression is similar to the voltage, the current can also be considered as sum of incident and reflected current waves.

The equations for voltage and currents can be rearranged as follows:

$$\begin{aligned} V &= V_r \cdot \frac{e^{\gamma x} + e^{-\gamma x}}{2} + I_r Z_c \frac{e^{\gamma x} - e^{-\gamma x}}{2} \\ &= V_r \cosh \gamma x + I_r Z_c \sinh \gamma x \end{aligned} \quad (4.38)$$

$$\begin{aligned} I &= \frac{1}{Z_c} \left[V_r \frac{e^{\gamma x} - e^{-\gamma x}}{2} + I_r Z_c \frac{e^{\gamma x} + e^{-\gamma x}}{2} \right] \\ &= \frac{1}{Z_c} [V_r \sinh \gamma x + I_r Z_c \cosh \gamma x] \\ &= \frac{V_r}{Z_c} \sinh \gamma x + I_r \cosh \gamma x \end{aligned} \quad (4.39)$$

Rewriting these equations for $x = l$, where $V = V_s$ and $I = I_s$

$$V_s = V_r \cosh \gamma l + I_r Z_c \sinh \gamma l \quad (4.40)$$

$$I_s = V_r \frac{\sinh \gamma l}{Z_c} + I_r \cosh \gamma l \quad (4.41)$$

These two equations relate the sending end voltage and current with the receiving end quantities. We have said previously that these quantities are related by the general equations.

$$V_s = AV_r + BI_r \quad (4.19)$$

$$I_s = CV_r + DI_r \quad (4.20)$$

where A , B , C and D are such that

$$A = D$$

and

$$AD - BC = 1$$

Comparing the coefficients of the equations (4.40) and (4.41) with equations (4.19) and (4.20) respectively,

$$A = \cosh \gamma l$$

$$B = Z_c \sinh \gamma l$$

$$C = \frac{\sinh \gamma l}{Z_c}$$

$$D = \cosh \gamma l$$

and

From this it is clear that

$$A = D = \cosh \gamma l$$

and

$$AD - BC = \cosh^2 \gamma l - Z_c \sinh \gamma l \cdot \frac{\sinh \gamma l}{Z_c} = 1.$$

Ferranti Effect

- When a long line is operating under no load or light load condition, the receiving end voltage is greater than the sending end voltage.
- This is known as Ferranti-effect.
- This phenomenon can be explained with the following reasonings:

(i) Assume no load condition. The equation (4.37)

$$V = \frac{V_r + I_r Z_c}{2} e^{\alpha x} e^{j\beta x} + \frac{V_r + I_r Z_c}{2} e^{-\alpha x} e^{-j\beta x}$$

reduces to:

$$V_s = \frac{V_r}{2} e^{\alpha l} e^{j\beta l} + \frac{V_r}{2} e^{-\alpha l} e^{-j\beta l}$$

when $x = l$ and $I_r = 0$.

At $l = 0$

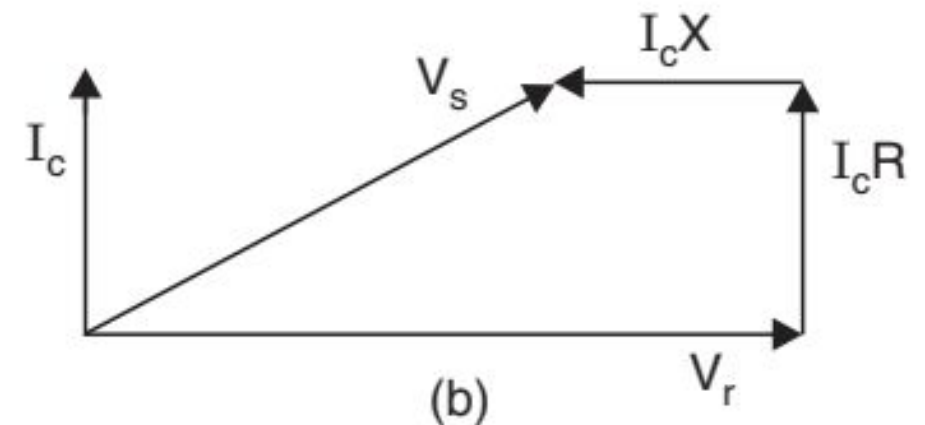
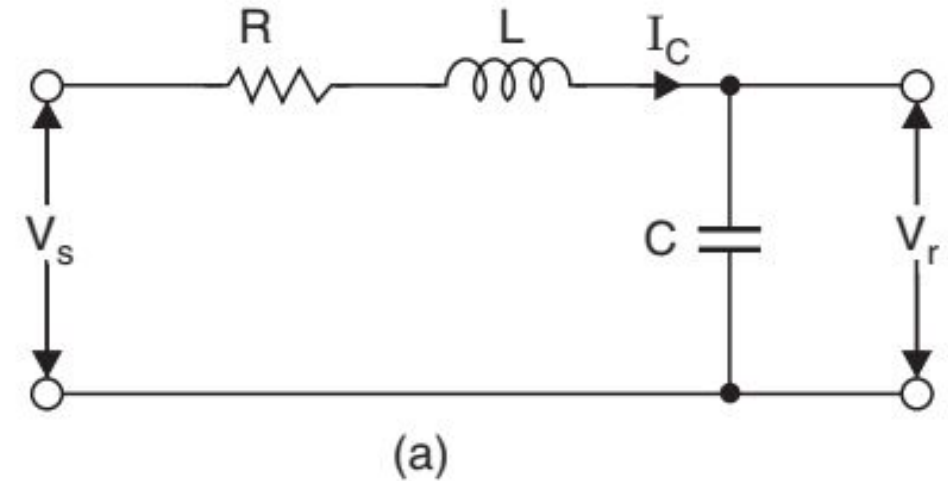
$$V_r = \frac{V_r}{2} + \frac{V_r}{2}$$

As l increases, the incident component of sending end voltage increases exponentially and turns the vector anti-clockwise through an angle βl , whereas the reflected part of sending end voltage decreases by the same amount and is rotated clockwise through the same angle βl .

The sum of these two components of sending end voltage gives a voltage which is smaller than V_r

(ii) A simple explanation of Ferranti-effect can be given by approximating the distributed parameters of the line by lumped impedance as shown in Fig. (a).

Since usually the capacitive reactance of the line is quite large as compared to the inductive reactance, under no load or lightly loaded condition the line current is of leading p.f. The phasor diagram is given below for this operating condition.



- The charging current produces drop in the reactance of the line which is in phase opposition to the receiving end voltage and hence the sending end voltage becomes smaller than the receiving end voltage.
- Yet another way of explaining the Ferranti-effect is based on the net reactive power flow on the line.
- It is known that if the reactive power generated at a point is more than the reactive power absorbed, the voltage at that point becomes higher than the normal value and vice versa.
- The inductive reactance of the line is a sink for the reactive power whereas the shunt capacitances generate reactive power.

- In fact, if the line loading corresponds to the surge impedance loading, the voltage is same everywhere as the reactive power absorbed then equals the reactive power generated by the line.
- The SIL, therefore, gives definite meaning to the terms lightly loaded or fully loaded lines.
- If the loading is less than SIL, the reactive power generated is more than absorbed, therefore, the receiving end voltage is greater than the sending end voltage.
- This explains, therefore, the phenomenon due to Ferranti-effect.

Power circle diagram

The performance of a transmission line can be studied either by analytical methods or by graphical methods. Purely analytical methods sometimes involve tedious calculations. They take considerable time and labour. Power circle diagrams provide a convenient graphical method for studying the performance of a transmission line under the conditions of the varying load. Circle diagrams are specially useful in system-stability studies, reactive volt-ampere calculations, and in the system design and operation. Circle diagrams are easy to construct and use. Their accuracy is fairly high.

12.2 RECEIVING-END VOLTAGE PHASOR DIAGRAM

The receiving-end phasor diagram for voltage is drawn with the help of the general equation

$$\mathbf{V}_s = \mathbf{A}\mathbf{V}_r + \mathbf{B}\mathbf{I}_r$$

The use of the above equation gives generalized result which may be applied to any specific case. The complex constants $\mathbf{A}$ and $\mathbf{B}$ may be written in the form of their magnitudes and arguments as

$$\mathbf{A} = A / \underline{\alpha}, \quad \mathbf{B} = B / \underline{\beta}$$

Let $\mathbf{V}_r$ be taken as the reference phasor. The receiving-end current $\mathbf{I}_r$ is assumed to have a phase difference of φ_r lagging. The constant $\mathbf{A}$ ($= \cosh \gamma S$) has a magnitude slightly less than unity. The argument α is a small positive angle. Angle β is nearly equal to 90° .

$$\mathbf{A}\mathbf{V}_r = A / \underline{\alpha} \cdot V_r / \underline{0^\circ} = AV_r / \underline{\alpha}$$

It indicates that the phasor $\mathbf{A}\mathbf{V}_r$ is leading $\mathbf{V}_r$ by angle α . Similarly, expressing $\mathbf{B}\mathbf{I}_r$ in the polar form as

$$\mathbf{B}\mathbf{I}_r = B / \underline{\beta} \cdot I_r / \underline{-\varphi_r} = BI_r / \underline{\beta - \varphi_r}$$

Thus, the phasor $\mathbf{B}\mathbf{I}_r$ leads $\mathbf{V}_r$ by an angle $(\beta - \varphi_r)$. $\mathbf{V}_s$ will be obtained by adding phasors $\mathbf{A}\mathbf{V}_r$ and $\mathbf{B}\mathbf{I}_r$. $\mathbf{V}_s$ leads $\mathbf{V}_r$ by an angle δ called the *torque angle*, *load angle* or *transmission angle*. The phasor diagram is shown in Fig. 12.1. $\mathbf{V}_s$ may be written as

$$\mathbf{V}_s = V_s / \underline{\delta}$$

δ changes with the change in load. Draw a line xx through O_r making an angle ϕ_r with phasor BI_r . The line $x_1O_1x_1$ is drawn parallel to $xO_r x$ and the line $yO_r y$ is drawn perpendicular to $xO_r x$. It is observed that the line $xO_r x$ makes an angle β with the reference phasor V_r .

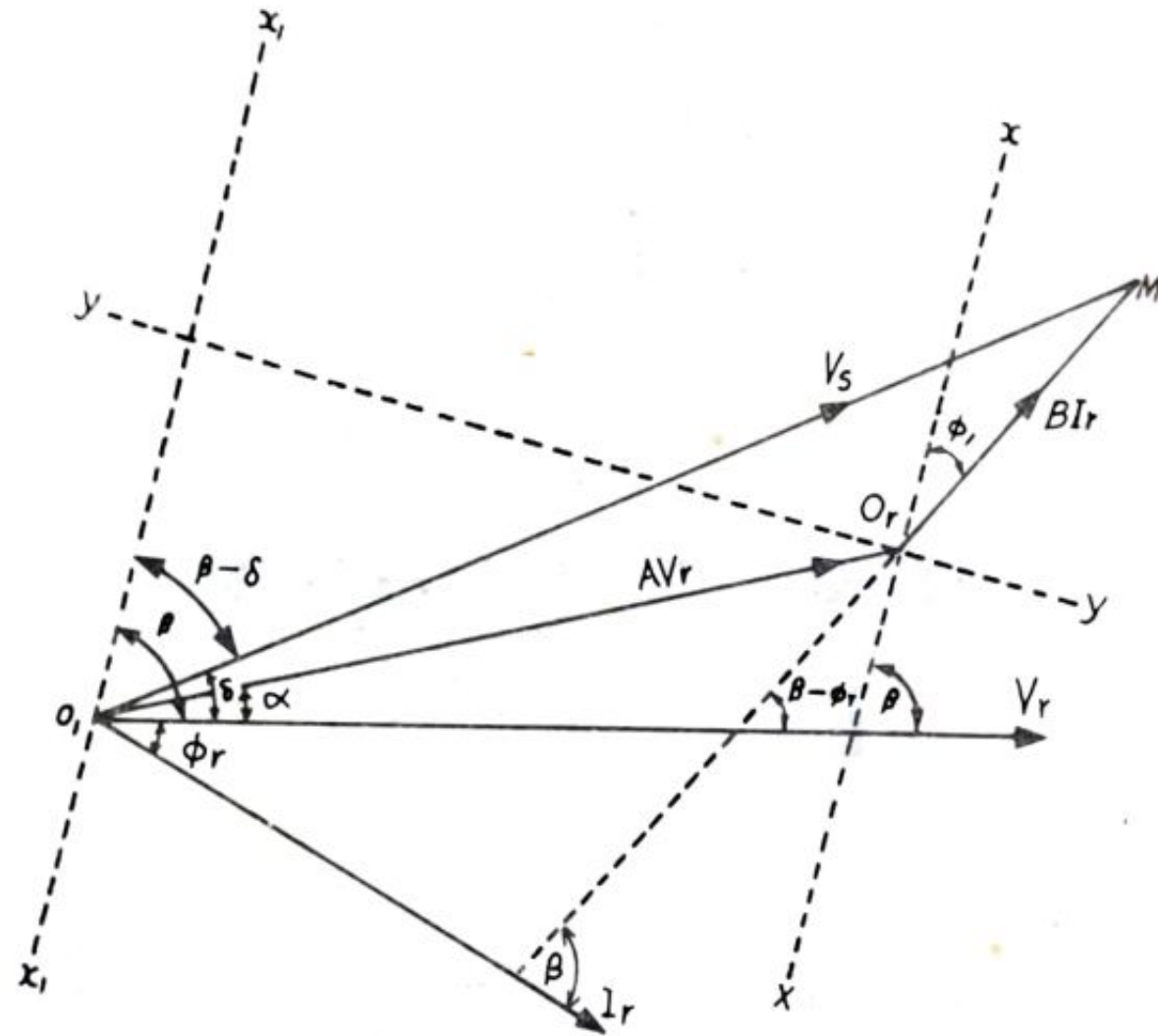


Fig. 12.1. Receiving-end voltage phasor diagram.

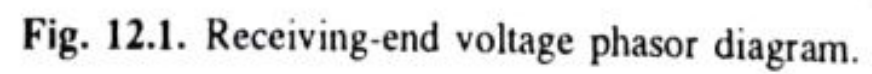


Fig. 12.1. Receiving-end voltage phasor diagram.

12.3 PHASOR VOLT-AMPERES

The phasor volt-amperes S is the product of phasor voltage and the conjugate of phasor current. The conjugate phasor has the same modulus but its angle is of the opposite sign.

Let the voltage at the receiving-end be taken as reference phasor. Then

$$\underline{V_r} = V_r / \underline{0^\circ}$$

For an inductive load the current lags the voltage. If ϕ_r be the phase angle of the current

$$\underline{I_r} = I_r / \underline{-\phi_r}$$

The conjugate of $\underline{I_r}$ will be denoted by $\underline{I_r^*}$. It is given by

$$\underline{I_r^*} = I_r / \underline{+\phi_r}$$

Phasor volt-amperes at the receiving-end of the line

$$\begin{aligned} S_r &= \mathbf{V}_r \cdot \mathbf{I}_r^* = V_r \angle 0^\circ \cdot I_r \angle +\varphi_r = V_r I_r \angle \varphi_r \\ &= V_r I_r \cos \varphi_r + j V_r I_r \sin \varphi_r = P_r + j Q_r \quad (\text{say}) \end{aligned} \quad (12.3.1)$$

The phasor product S_r so obtained is the sum of two components. The real part, denoted by P_r is called the *active or real power*. The imaginary part Q_r is known as *reactive volt- amperes*. P_r is expressed in watts, kilowatts, or megawatts. The quadrature component Q_r is expressed in vars (from the initials of volt amperes reactive), kilovars or megavars.

For a capacitive load for which the power factor is leading, the phasor volt amperes S_r is given by

$$\begin{aligned} S_r &= \mathbf{V}_r \cdot \mathbf{I}_r^* = V_r \angle 0^\circ \cdot I_r \angle -\varphi_r = V_r I_r \angle -\varphi_r \\ &= V_r I_r \cos \varphi_r - j V_r I_r \sin \varphi_r = P_r - j Q_r \end{aligned} \quad (12.3.2)$$

Q_r obtained in this case is negative. We shall adopt the sign convention that the volt amperes reactive taken by an inductive load are positive and by a capacitive load negative. The signs of Q_r obtained in Eqs. (12.3.1) and (12.3.2) agree with the convention of sign adopted by us.

The phasor volt-amperes $\mathbf{S}$ may also be found by multiplying the phasor current and the conjugate of the phasor voltage, but the convention adopted by us is more generally accepted.

12.4 RECEIVING-END POWER CIRCLE DIAGRAM

Receiving-end power circle diagram may be derived from graphical or analytical methods. First we shall draw the circle diagram based upon the graphical considerations. The receiving-end power circle diagram may be developed from the voltage phasor diagram of Fig. 12.1. Each phasor on this diagram is multiplied by V_r/B . The results of the multiplication are shown as follows :

$$\mathbf{V}_r \cdot \frac{\mathbf{V}_r}{B} = \frac{V_r^2}{B} \underline{-\beta}$$

$$\mathbf{A} \mathbf{V}_r \cdot \frac{\mathbf{V}_r}{\mathbf{B}} = \frac{A V_r^2}{B} / \underline{\alpha - \beta}$$

$$\mathbf{B} \mathbf{I}_r \cdot \frac{\mathbf{V}_r}{\mathbf{B}} = V_r I_r / -\underline{\Phi_r}$$

$$\mathbf{V}_s \cdot \frac{\mathbf{V}_r}{\mathbf{B}} = \frac{V_s V_r}{B} / \delta - \beta$$

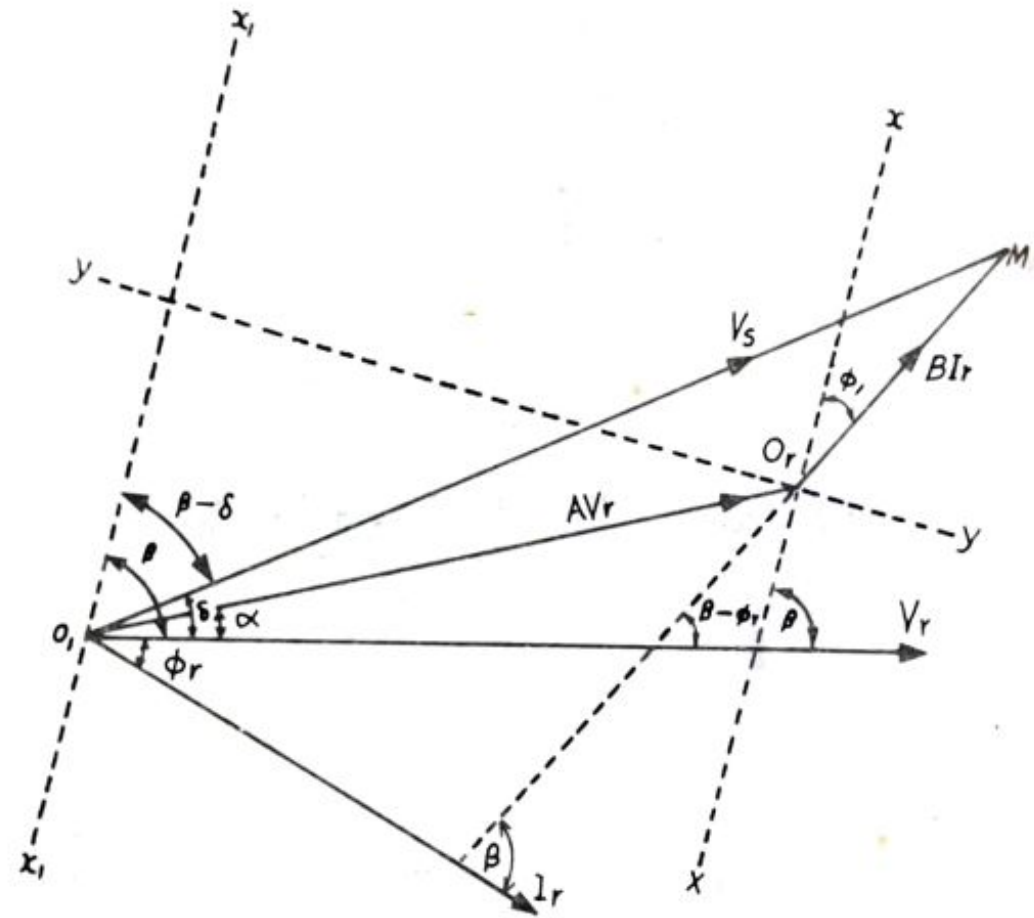


Fig. 12.1. Receiving-end voltage phasor diagram.

By such a multiplication all the voltage phasors of Fig. 12.1 are converted into volt-amperes and are rotated through an angle $-\beta$. The diagram so obtained is shown in Fig. 12.2.

Let the line $xO_r x$ be taken as horizontal axis, $yO_r y$ the vertical axis, and O_r the origin of coordinates. In Fig. 12.2 the product $V_r I_r$ gives the total volt-amperes at the receiving-end. The horizontal projection of $V_r I_r$ is $V_r I_r \cos \phi_r$ which is the real power. The vertical projection of $V_r I_r$ is $V_r I_r \sin \phi_r$ which is the volt-amperes reactive. For this reason, the $xO_r x$ and $yO_r y$ may be marked as real power and reactive volt-amperes axes respectively.

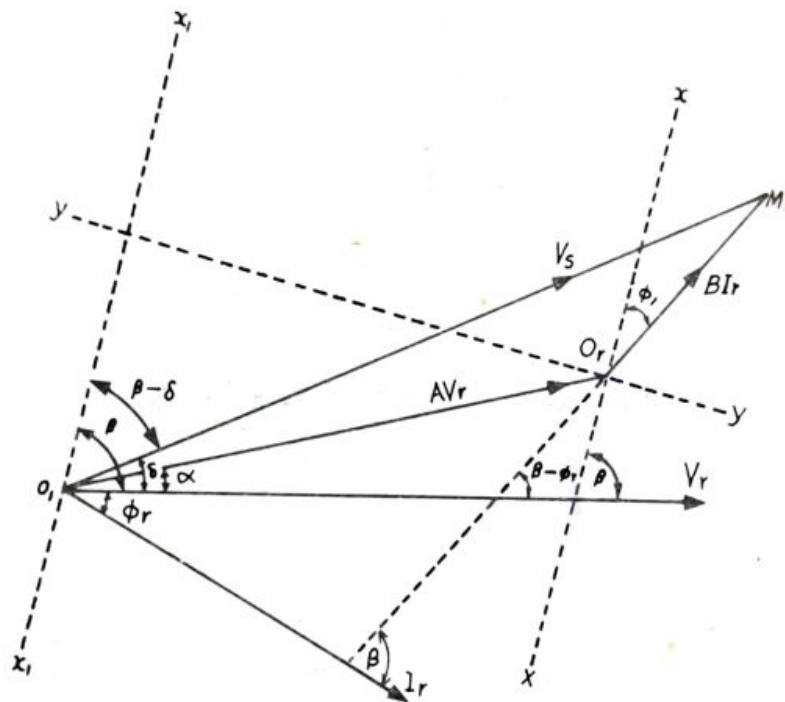


Fig. 12.1. Receiving-end voltage phasor diagram.

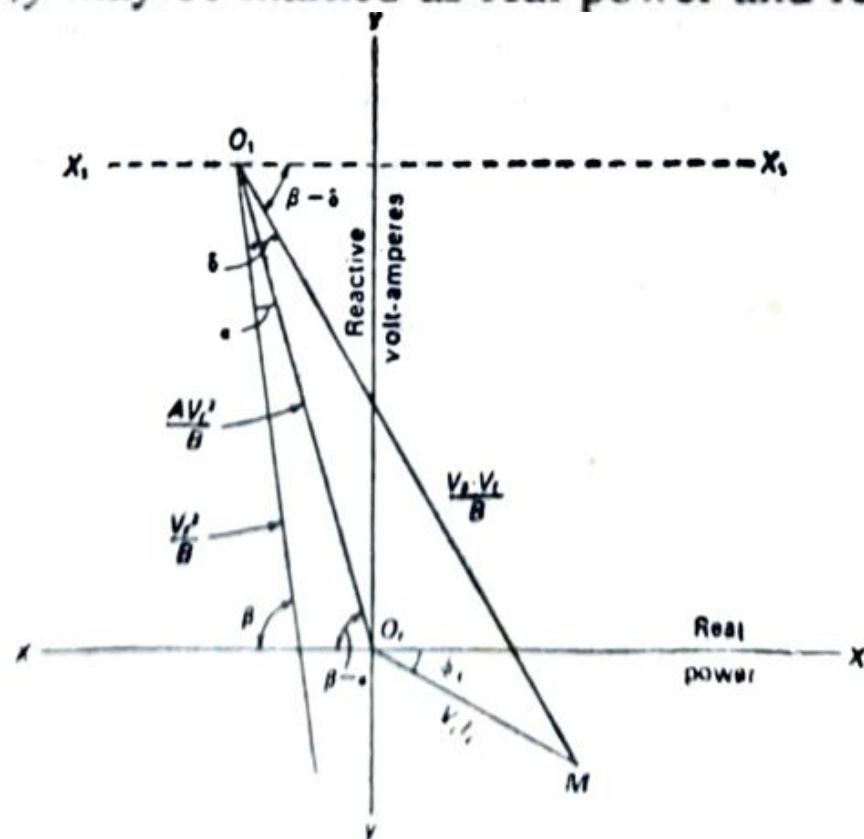


Fig. 12.2.

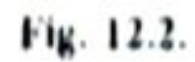


Fig. 12.2.

Under the sign convention adopted that an inductive load draws positive reactive volt-amperes, the diagram shown in Fig. 12.2 should be rotated about the horizontal axis. This gives the diagram shown in Fig. 12.3.

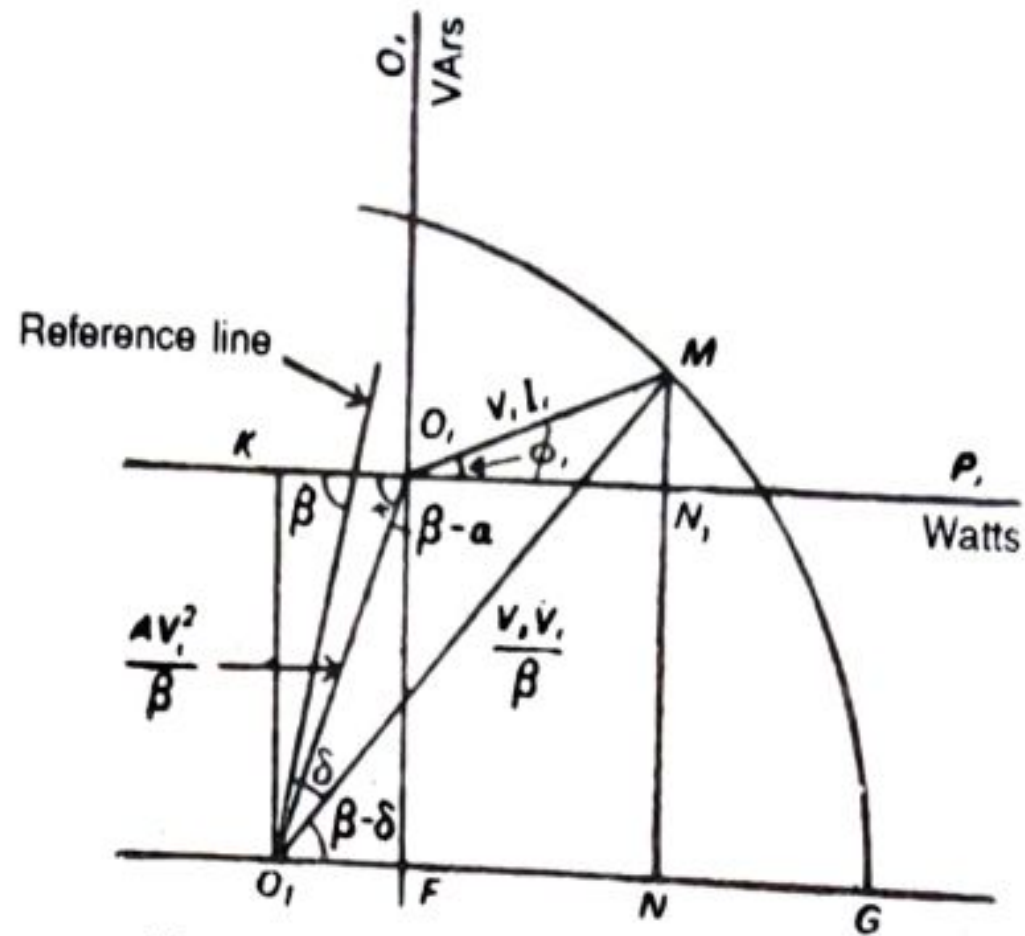


Fig. 12.3. Receiving-end power circle diagram.

The following facts are derived from Fig. 12.3. If the sending-end and receiving-end voltages V_s and V_r are kept constant the position of the point O_1 does not change as the distance $O_1O_r \left(= \frac{AV_r^2}{B} \right)$ is a constant and is independent of current I_r . Also the distance of the operating point M from O_1 , i.e., $\frac{V_s V_r}{B}$ is constant for fixed values of V_s and V_r . With the variation of the load I_r and ϕ_r , the position of the point M and therefore, the distance $O_rM (= V_r I_r)$ will change, but the point M will remain at a constant distance $O_1M \left(= \frac{V_s V_r}{B} \right)$ from the fixed point O_1 . Thus, M will move along a circle of radius $\frac{V_s V_r}{B}$ with its centre at O_1 . Since the locus of the operating point M is a circle with the axis of reference as real power, and reactive voltamperes at the receiving-end, the diagram so obtained is called the *receiving-end power circle diagram*.

The coordinates of the centre of the receiving-end power circle diagram referred to a system of rectangular coordinates in which the abscissae represent the real power P_r , and the ordinates represent the reactive volt-amperes Q_r are $(-O_r K, -KO_1)$ in Fig. 12.3.

$$-O_r K = -\frac{A}{B} V_r^2 \cos(\beta - \alpha) \text{ watts}$$

$$-KO_1 = -\frac{A}{B} V_r^2 \sin(\beta - \alpha) \text{ vars}$$

$$\text{Radius of a receiving-end power circle, } \rho_r = O_1 M = \frac{V_s V_r}{B} \text{ volt-amperes}$$

The distance of the centre O_1 from the origin O_r is

$$O_r O_1 = \frac{A}{B} V_r^2$$

The equation of the power circle referred to P_r and Q_r as reference axes, centre $(-O_r K, -KO_1)$ and radius $O_1 M$ may be written as

$$(P_r + O_r K)^2 + (Q_r + KO_1)^2 = (O_1 M)^2$$

ordinates represent

(12.4.1)

Substituting the values obtained from Eq. (12.4.1) in the above equation, we get

$$\left[P_r + \frac{A}{B} V_r^2 \cos (\beta - \alpha) \right]^2 + \left[Q_r + \frac{A}{B} V_r^2 \sin (\beta - \alpha) \right]^2 = \left(\frac{V_s V_r}{B} \right)^2 \quad (12.4.2)$$

It is to be noted that line $\frac{V_r^2}{B}$ in Fig. 12.3 is marked as *reference line* from which torque angle δ is measured. This line in Fig. 12.3 corresponds to phasor V_r , which was originally taken as reference phasor in Fig. 12.1. In Fig. 12.3, δ is the angle between the line $\frac{V_r^2}{B}$ and the line $O_1 M$.

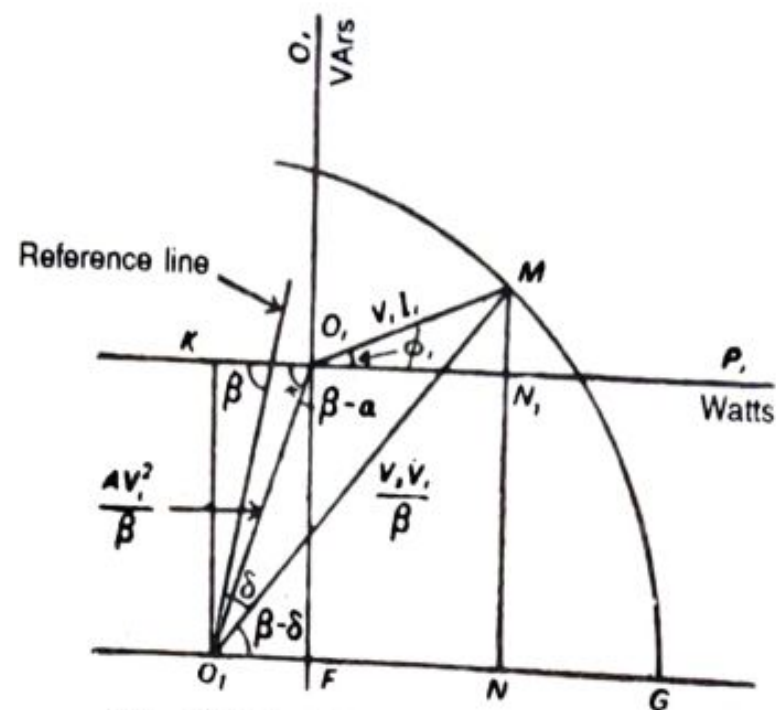


Fig. 12.3. Receiving-end power circle diagram.

We shall consider two cases. In the first case, suppose that V_r is fixed and it is desired to draw receiving-end power circle diagrams for different values of V_s . The position of the centre O_1 , which is independent of V_s , is unchanged while the radii of the circles will be different. In other words, we get the circles having the same centre O_1 , but different radii corresponding to different values of V_s . Such concentric circles for a fixed value of V_r and different value of V_s are shown in Fig. 12.4.

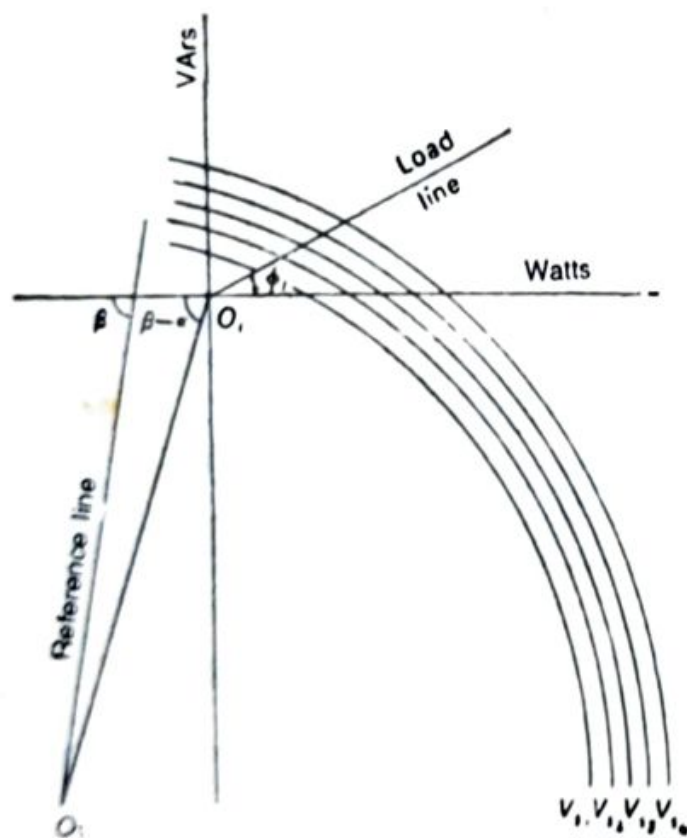


Fig. 12.4. Receiving-end power circles for different values of V_s and a fixed value of V_r .

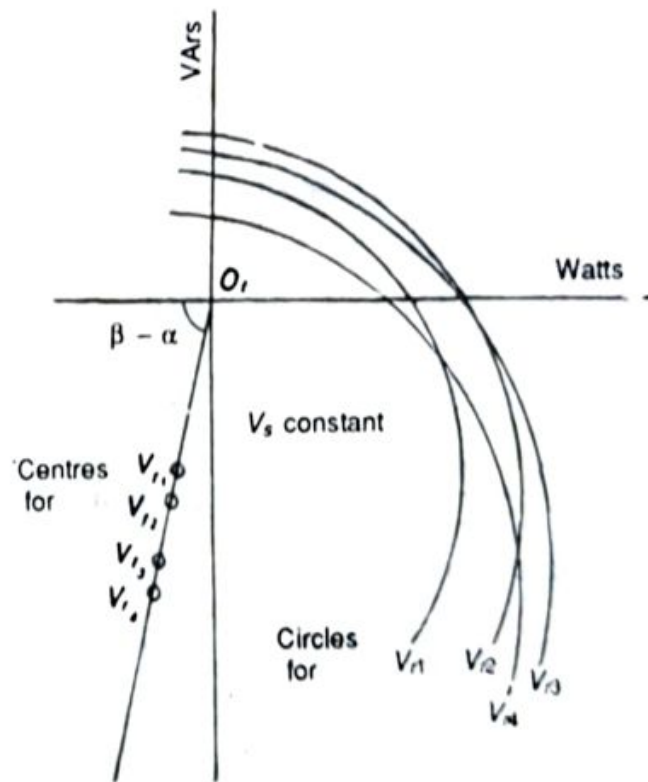


Fig. 12.5.

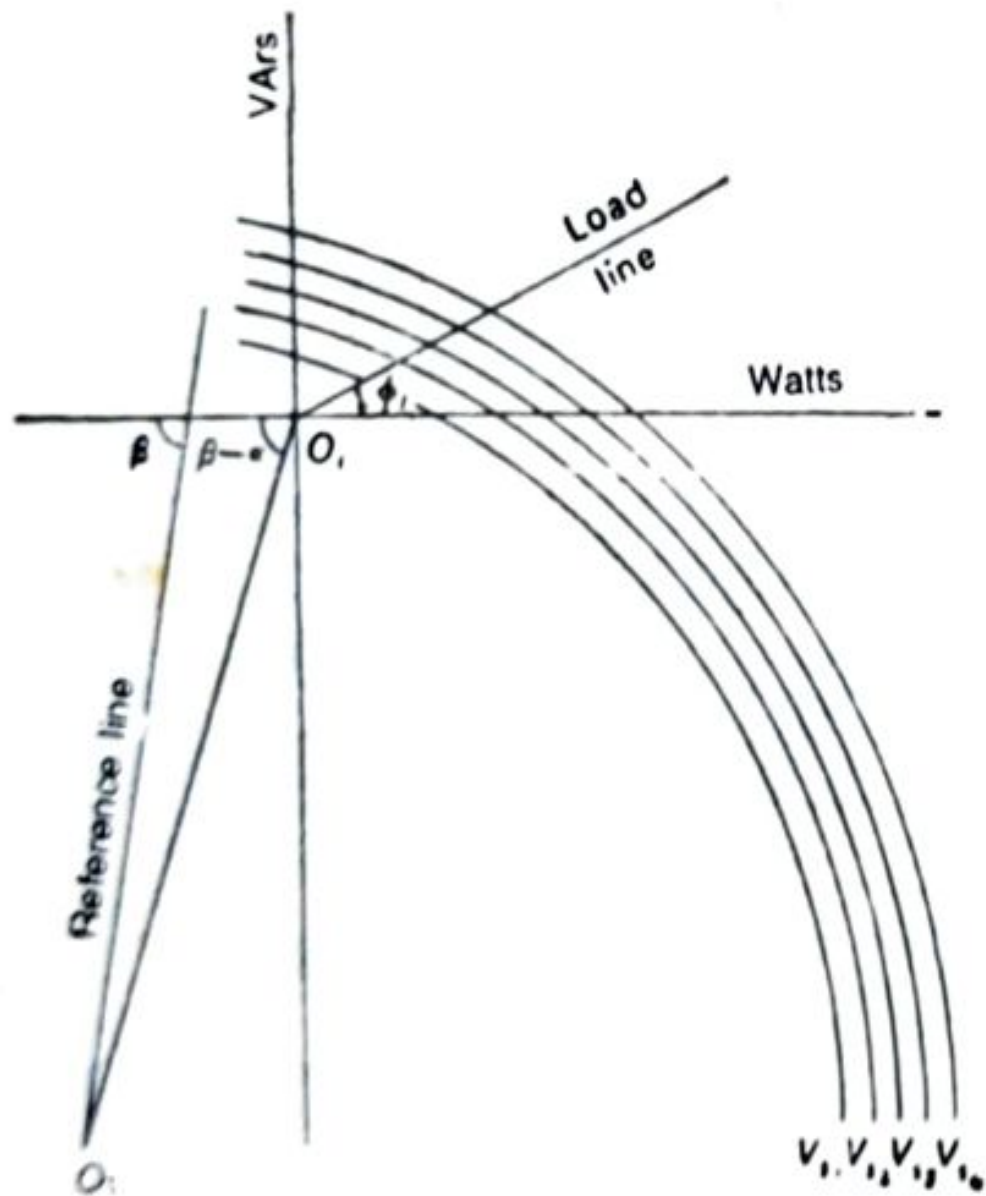


Fig. 12.4. Receiving-end power circles for different values of V_r , and a fixed value of V_s .

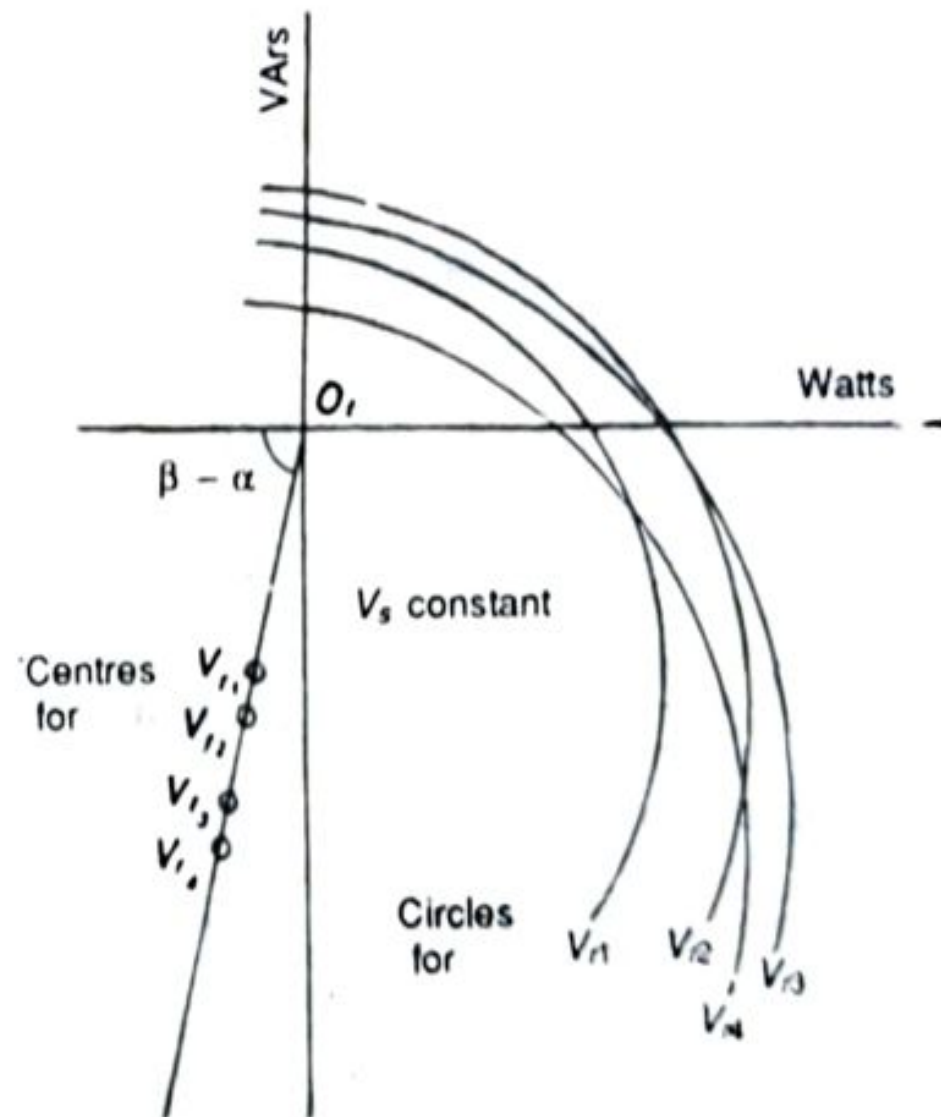


Fig. 12.5.

Secondly, let V_s be kept constant and it is desired to draw receiving-end power circles for different values of V_r . The circles will not be concentric. The positions of the centres will be different for different values of V_r , but they shall be on the line through the origin and making an angle $(\beta - \alpha)$ with the horizontal (Fig. 12.5).

If V_s and V_r are expressed in phase volts, i.e., line to neutral volts, the coordinates of the diagram given in Fig. 12.3 will represent watts per phase and vars per phase respectively. Each length on the diagram is determined by the product of two voltages or the square of the voltage. For a balanced three-phase circuit the line-to-line voltage is $\sqrt{3}$ times the phase voltage. Thus, if V_s and V_r are expressed in line-to-line volts, each length on the diagram becomes three times its value when V_s and V_r are expressed in phase volts. The coordinates in this case will, therefore, represent the total three-phase watts and vars respectively. If the voltages, V_s and V_r are expressed in line-to-line kilovolts the coordinates of the diagram will represent megawatts and megavars respectively for all the three phases.

Thus, if in Eq. (12.4.1) the voltages are in line-to-line kilovolts,

$$\left. \begin{aligned} \text{abscissa of the centre of the receiving-end circle} &= -\frac{A}{B} V_r^2 \cos (\beta - \alpha) \text{ MW} \\ \text{ordinate of the centre of the receiving-end circle} &= -\frac{A}{B} V_r^2 \sin (\beta - \alpha) \text{ MVAr} \\ \text{radius of the receiving-end circle} &= \frac{V_s V_r}{B} \text{ MVA} \end{aligned} \right\} \quad (12.4.3)$$

12.6 SENDING-END VOLTAGE PHASOR DIAGRAM

The basic voltage phasor diagram may be constructed from the equation

$$\mathbf{V}_r = \mathbf{D}\mathbf{V}_s - \mathbf{B}\mathbf{I}_s$$

Let us take $\mathbf{V}_s$ as the reference phasor so that

$$\mathbf{V}_s = V_s \angle 0^\circ$$

$$\mathbf{D}\mathbf{V}_s = D \angle \Delta \cdot V_s \angle 0^\circ = DV_s \angle \Delta \quad \checkmark$$

Thus, the phasor $\mathbf{D}\mathbf{V}_s$ leads the phasor $\mathbf{V}_s$ by an angle Δ .

$$\mathbf{B}\mathbf{I}_s = B \angle \beta I_s \angle -\varphi_s = BI_s \angle \beta - \varphi_s$$

The phasor $\mathbf{B}\mathbf{I}_s$ leads the phasor $\mathbf{V}_s$ by an angle $(\beta - \varphi_s)$. The receiving-end voltage $\mathbf{V}_r$ is obtained by subtracting the phasor $\mathbf{B}\mathbf{I}_s$ from the phasor $\mathbf{D}\mathbf{V}_s$.

$$\mathbf{V}_r = V_r \angle -\delta$$

The sending-end voltage phasor diagram is shown in Fig. 12.6.

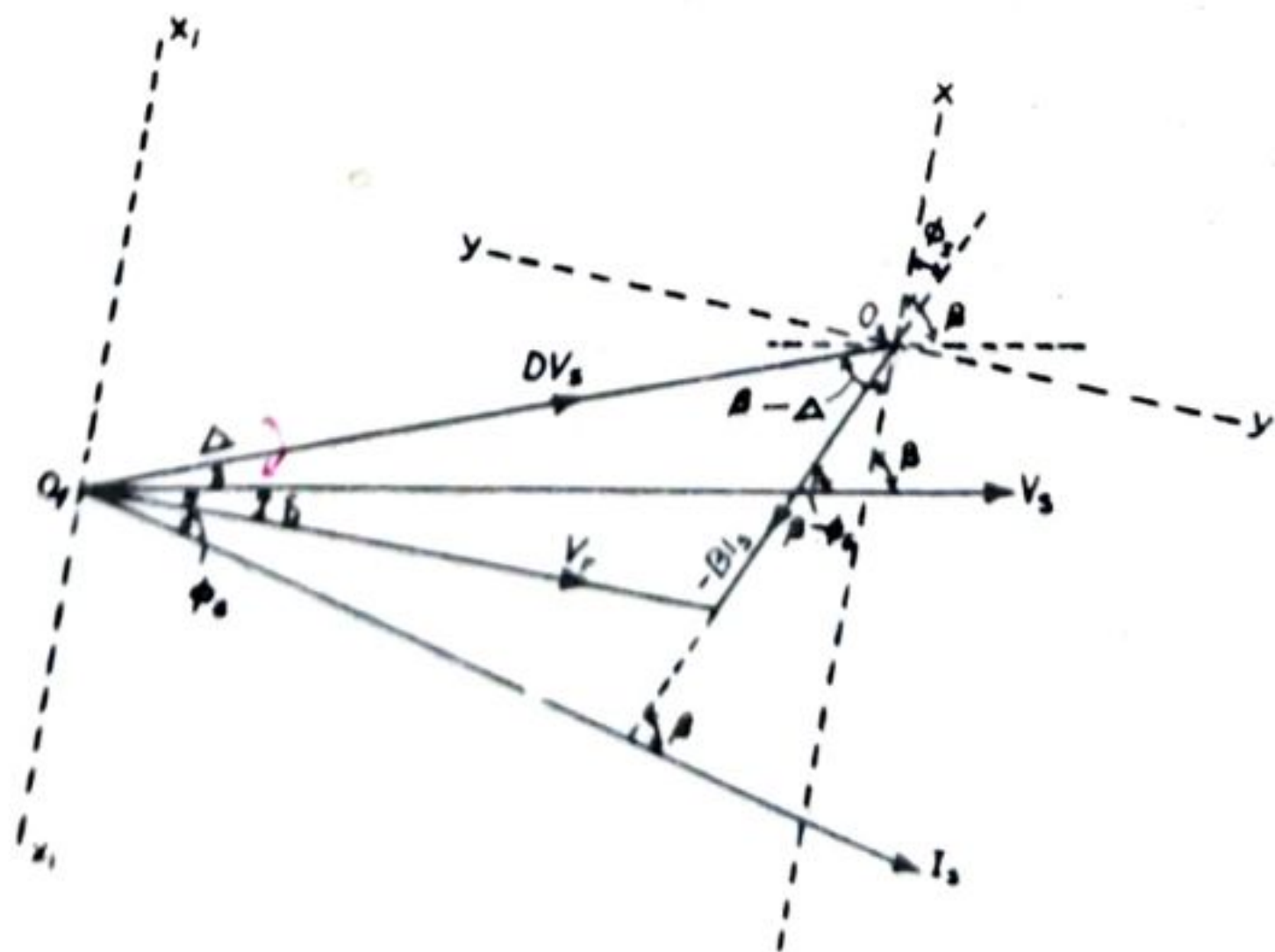


Fig. 12.6. Sending-end voltage phasor diagram.

The sending-end voltage phasor diagram is shown in Fig. 12.6.

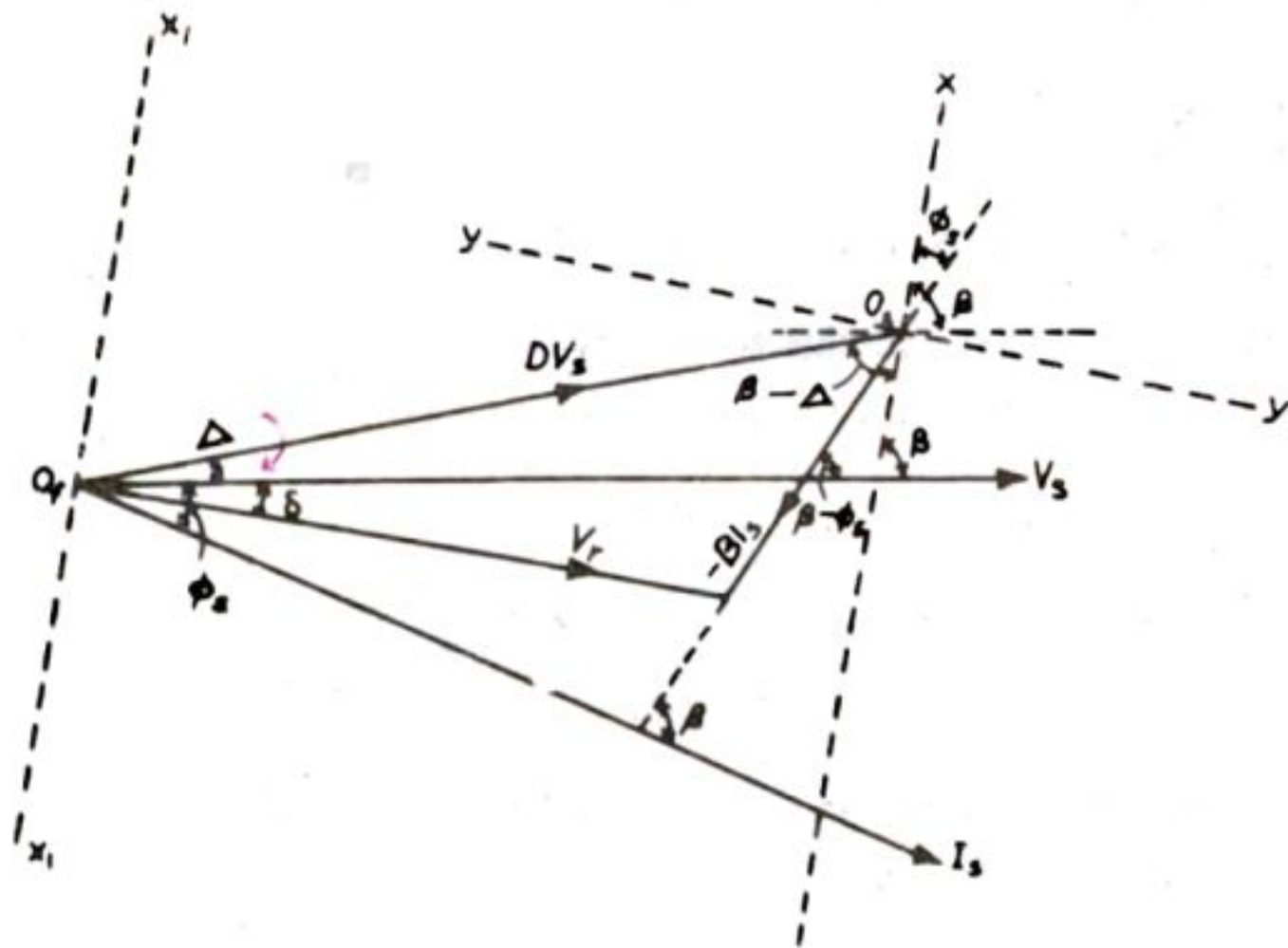


Fig. 12.6. Sending-end voltage phasor diagram.

12.7 SENDING-END POWER CIRCLE DIAGRAM

The sending-end power circle is drawn graphically from the voltage phasor diagram at the sending-end in exactly the same manner in which the receiving-end power circle is drawn.

Each phasor in Fig. 12.6 is multiplied by $\frac{-V_s}{B}$

$$\frac{-V_s}{B} = \frac{V_s / 180^\circ}{B / \beta} = \frac{V_s}{B} / 180^\circ - \beta \quad \checkmark$$

$$\frac{V_r (-V_s)}{B} = V_r / -\delta \cdot \frac{V_s / 180^\circ}{B / \beta} = \frac{V_s V_r}{B} / 180^\circ - \beta - \delta$$

$$\frac{-V_s}{B} \cdot DV_s = \frac{DV_s^2}{B} / \Delta + 180^\circ - \beta$$

$$(-BL_s) \left(\frac{-V_s}{B} \right) = V_s I_s \quad \checkmark$$

If V_s and V_r are kept constant the position of point O_1 becomes fixed. The distance of the operating point M from O_1 , i.e., $O_1 M \left(= \frac{V_s V_r}{B} \right)$ is also a constant for the fixed values of V_s and V_r . Any variation in load will change the position of operating point M , but it shall remain fixed at a distance $O_1 M \left(= \frac{V_s V_r}{B} \right)$ from the fixed point O_1 . Thus, the locus of the operating point M is a circle with centre O_1 and radius $O_1 M \left(= \frac{V_s V_r}{B} \right)$. Since the locus of the operating point is a circle with the axes of reference as active power and reactive volt-amperes at the sending-end, the diagram so obtained is called the *sending-end power circle diagram*.

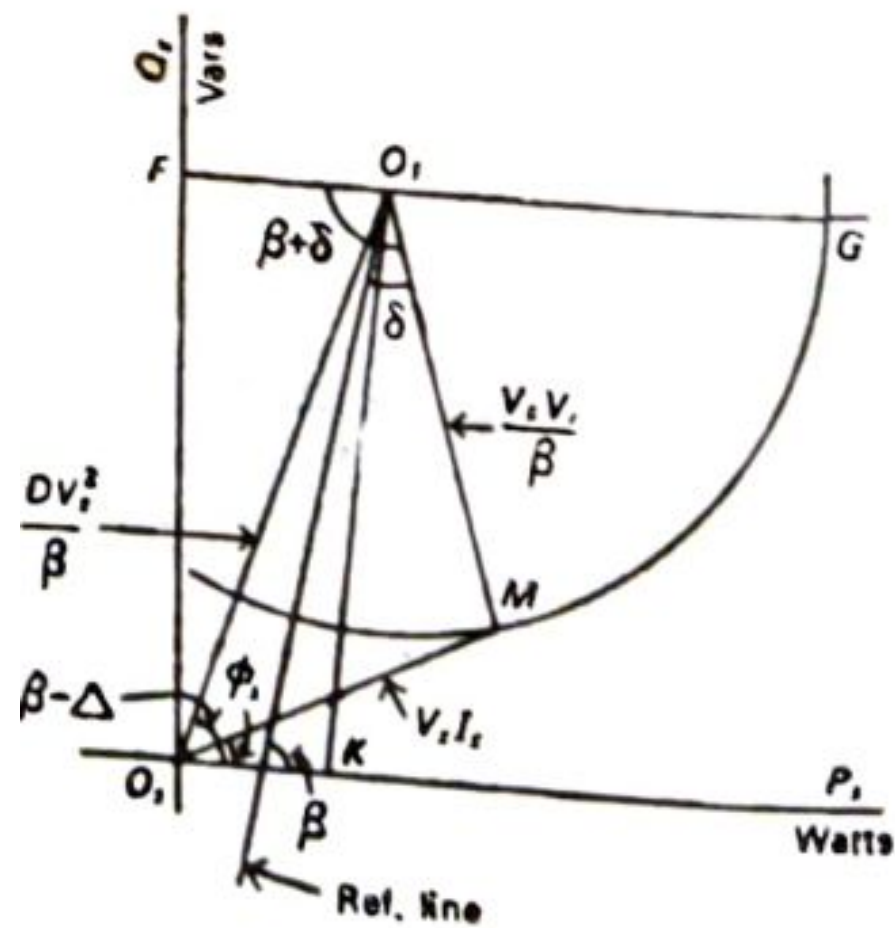


Fig. 12.8. Sending-end power circle diagram.

The coordinates of the centre of the sending-end power circle diagram referred to a system of rectangular coordinates

in which the abscissae represent the real power P_s and the ordinates represent the reactive volt-amperes Q_s at the sending-end are $(O_s K, KO_1)$ in Fig. 12.8.

$$\left. \begin{aligned} \text{Abcissa, } O_s K &= \frac{DV_s^2}{B} \cos (\beta - \Delta) \text{ watts} \\ \text{Ordinate, } KO_1 &= \frac{DV_s^2}{B} \sin (\beta - \Delta) \text{ vars} \\ \text{Radius of the sending-end power circle, } \rho_s = O_1 M &= \frac{V_s V_r}{B} \text{ volt-amperes} \end{aligned} \right\} \quad (12.7.1)$$

The distance of the centre O_1 from origin O_s is

$$O_s O_1 = \frac{DV_s^2}{B}$$

The equation of the power circle referred to P_s and Q_s as reference axes, centre $(O_s K, KO_1)$ and radius $O_1 M$ may be written as

$$(P_s - O_s K)^2 + (Q_s - KO_1)^2 = (O_1 M)^2$$

Substituting the values obtained from Eq. (12.7.1) in the above equation, we obtain

$$\left[P_s - \frac{DV_s^2}{B} \cos(\beta - \Delta) \right]^2 + \left[Q_s - \frac{DV_s^2}{B} \sin(\beta - \Delta) \right]^2 = \left(\frac{V_s V_r}{B} \right)^2 \quad (12.7.2)$$

$\frac{V_s^2}{B}$ is the *reference line* from which torque angle δ is measured on the sending-end power circle diagram.

Diagram:

If V_s be maintained constant then we get a family of concentric circles for various values of V_r . If V_r be kept constant then again we get a family of circles for various values of V_s , but these circles are not concentric. The centres of the circle all lie on the same line making an angle $(\beta - \Delta)$ with the horizontal.

3.24 SYNCHRONOUS PHASE MODIFIERS

It is well-known that a synchronous machine can be made to take either a lagging or a leading current from the line depending upon whether it is under-or over-excited. A synchronous motor used for taking only a leading current (say, for power factor improvement at the receiving end of a distribution system is referred to as a synchronous capacitor).

However, an idle running synchronous machine may be used for voltage regulation by connecting it in parallel with load at the receiving end of the line. Such an application may require the motor to take either a leading or lagging current (depending upon load conditions). When used in this manner it is more appropriately referred to as *synchronous phase modifier (SPM)*.

Phase modifiers differ from ordinary synchronous machines in as much as these are (unlike the latter) not meant for driving mechanical load and are built for highest possible economic speeds. These are equipped with smaller shafts and bearings, a high overall efficiency being specially aimed at. Standard machines for this purpose are designed to give full load (rated) output at leading power factor and can carry 50% of their capacity at lagging p.f.

Example 3.27. A 3-phase transmission line, 160 km long, transmits a load of 90,000 kW at 0.80 p.f. lagging. The line

voltage at the receiving end is 230 kV. The constants of the line are as follows :

$$A = D = 0.9785 \angle 0.3^\circ$$

$$B = 852 \angle 77.47^\circ$$

$$C = 0.000503 \angle 90.1^\circ$$

Construct the receiving end and sending end circle diagrams for the transmission line and calculate

- (a) sending end voltage, current factor, regulation and efficiency of the transmission line
- (b) load in kW at 0.80 power factor lagging that could be carried at 8% regulation
- (c) voltage drop if the load is 120 MW at the same power factor and the MVAR leading required for 120 MW load at 8% regulation.
- (d) rating and power factor of the SPM connected in parallel with the load so that the voltage transmitted at the sending end (230 kV) is the same as at receiving end.
- (e) what will be the maximum power that can be transmitted in case (d) ? Also calculate the corresponding p.f.

Solution. $P = 90,000 \text{ kW}$

$$V_r = \frac{230}{\sqrt{3}} \text{ kV}$$

p.f. = 0.85 lagging

$$I_r = \frac{90 \times 10^3}{\sqrt{3} \times 230 \times 0.85} = 268 \text{ A}$$

$$\angle(\beta - \alpha) = 77.47^\circ - 0.3^\circ = 77.17^\circ$$

and

Scales chosen,

$$20 \text{ MW} = 1 \text{ cm (horizontal)}$$

$$20 \text{ MVAR} = 1 \text{ cm (vertical)}$$

$$\text{So that } 20 \text{ MVA} = 1 \text{ cm}$$

Receiving end circle diagram (Fig. E3.22)

Horizontal co-ordinate,

$$= -\left| \frac{A}{B} \right| V_R^2 \cos(\beta - \alpha)$$

$$= -\frac{0.9785}{85.2} \times \left(\frac{230}{\sqrt{3}} \right)^2 \times 0.222$$

$$= -45 \text{ MW}$$

Vertical co-ordinate,

$$= -\left| \frac{A}{B} \right| V_R^2 \sin(\beta - \alpha)$$

$$= -\frac{0.9785}{85.2} \times \left(\frac{230}{\sqrt{3}} \right)^2 \times 0.975$$

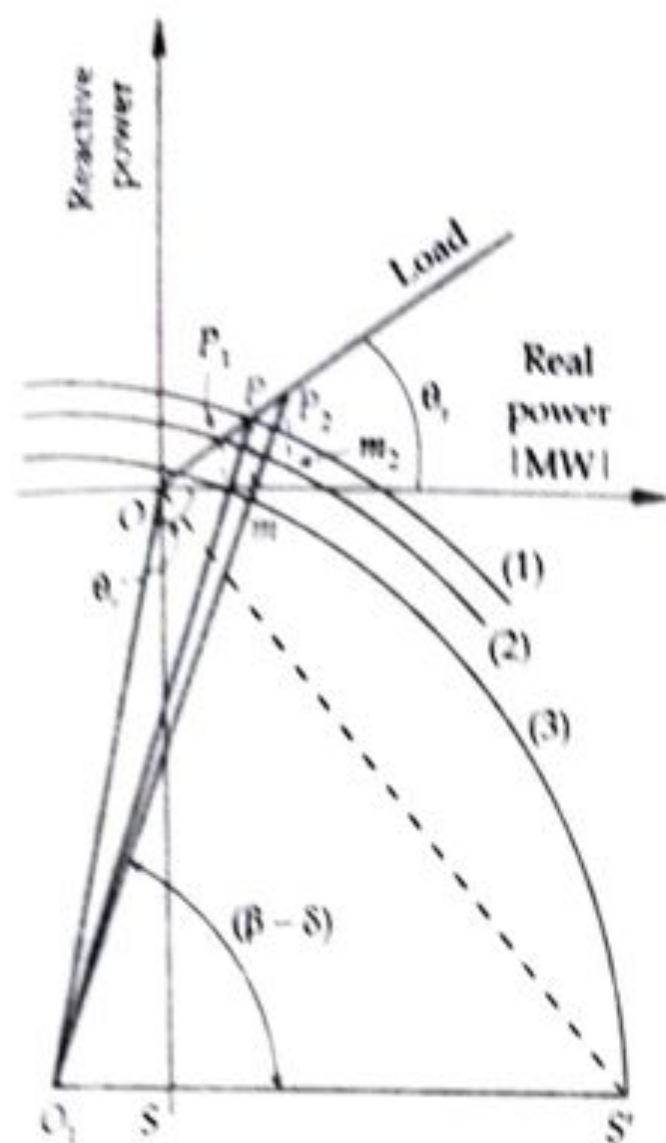
$$= -197 \text{ MVAR}$$

remembering that torque angle for both must be the same and that the two points must lie on the same $|V_s||V_r|$ circle.

Steps for drawing :

1. Locate point O_1 having vertical and horizontal ordinates as -197 and -45.
2. Draw load line at an angle $\theta_r = \cos^{-1} 0.80 = 36.7^\circ$ to the horizontal and cut.

$$\begin{aligned}
 OP &= |V_r||I_r| \\
 &= \frac{230}{\sqrt{3}} \times 268 \times 10^{-3} \\
 &= 35.6 \text{ MW}
 \end{aligned}$$



3. From the graph,

$$O_1P = 11.5 \text{ cm} \\ = 230 \text{ MVA}$$

$$\text{But } O_1P = \frac{|V_s||V_r|}{|B|}$$

$$\therefore |V_s| = \frac{230 \times 85.2}{230/\sqrt{3}} \\ = 147.6 \text{ kV}$$

$$\text{Line-to-line sending end voltage} \\ = \sqrt{3} \times 147.6 \\ = 256 \text{ kV}$$

4. $\beta - \delta = 71^\circ$ (from graph)

$$\therefore \delta = 77.47^\circ - 71^\circ \\ = 6.47^\circ$$

Sending end circle diagram (Fig. E.3.23)

Horizontal co-ordinate

$$= \frac{|D|}{|B|} |V_s|^2 \cos(\beta - \Delta) \\ = \frac{0.9785}{85.2} \left(\frac{256}{\sqrt{3}} \right)^2 \times 0.222 \\ = -55.6 \text{ MW}$$

Vertical co-ordinate

$$= \frac{|D|}{|B|} |V_s|^2 \sin(\beta - \Delta) \\ = \frac{0.9785}{85.2} \times \left(\frac{256}{\sqrt{3}} \right)^2 \times 0.975 \\ = 246 \text{ MVAR}$$

$$\angle \beta + \delta = 77.47^\circ + 6.47^\circ = 83.94^\circ$$

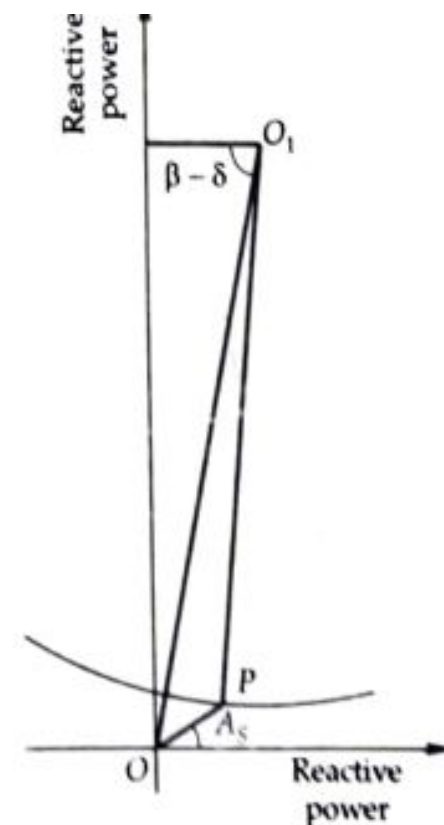


Fig. E3.23

Steps for drawing :

1. Again locate point O_1 having above coordinates

2. Draw circle diagram with radius = $\frac{|V_S||V_R|}{|B|}$

$$= \frac{(256 / \sqrt{3})(230 / \sqrt{3})}{852}$$

$$= 230 \text{ MVA} = 11.5 \text{ cm}$$

3. Draw $O_1 P$ at an angle $(\beta + \delta)$

4. Join OP

(a) (i) Sending end voltage = 256 kV (line-to-line)

(ii) Sending end current (I_s)

From sending end circle diagram

$$OP = |V_s||I_s|$$

But $OP = 1.8 \text{ cm} = 36 \text{ MVA}$ (measured)

$$\therefore |I_s| = \frac{36 \times 10^6}{(256 / \sqrt{3}) \times 10^3} = 244 \text{ amps}$$

(iii) Sending end power factor is $\cos \theta_s$

$$\theta_s = 27^\circ \text{ (measured)}$$

$$\therefore \cos \theta_s = 0.89$$

$$(iv) \text{ Regulation} = \frac{256 - 230}{230} \times 100 = 11.3\%$$

(v) Power at the sending end

$$\begin{aligned} &= 3|V_s||I_s|\cos\theta_s \\ &= 3 \times 147.6 \times 10^{-3} \times 244 \times 0.89 \\ &= 96 \text{ MW} \end{aligned}$$

$\therefore$ Transmission efficiency

$$= \frac{90.0}{96} \times 100 = 93.8\%$$

(b) A regulation of 8% means that

$$S = \left(\frac{V_s}{V_r} - 1 \right) \times 100$$

$$V_s = 1.08 V_r = 1.08 \left(\frac{230}{\sqrt{3}} \right)$$

or

$\therefore$ Radius of power circle diagram

$$= \frac{|V_s||V_r|}{|B|} = 1.08 \times \frac{(230/\sqrt{3})(230/\sqrt{3})}{85.2}$$

$$= 224 \text{ MVA} = 11.2 \text{ cm}$$

Draw receiving end circle diagram (2) corresponding to this cutting load line at P_1 . Horizontal component of $OP_1 = 1.25 \text{ cm} = 25 \text{ MW}$ which is load carried by line per phase.

$\therefore$ Total load carried at 8% regulation

$$= 25 \times 3 = 75 \text{ MW}$$

$$(c) \text{ Power per phase} = \frac{120}{3} = 40 \text{ MW}$$

The new point P_2 can be marked on the load line corresponding to this load of 40 MW.

$$O_1 P_2 = 119 \text{ cm}$$

$$= 238 \text{ MVA} = \frac{|V_s| |V_R|}{|B|}$$

$$\therefore |V_s| = \frac{238 \times 85.2}{(230/\sqrt{3})} = 152.5 \text{ kV}$$

$$\therefore \text{Drop per phase} = 152.5 - 133 = 19.5 \text{ kV}$$

Next since the regulation required is 8%

$$V_r = 108 \text{ V}$$

For the increase load, therefore, P_2 must move vertically downward to cut circle diagram (2) at u .
leading MVAR = $P_2 m_2 = 0.75 \text{ cm} = 16 \text{ MVAR per phase}$
or Total MVAR required = $16 \times 3 = 48 \text{ MVAR}$

(d) In this case $|V_s| = |V_r|$

Radius of receiving end circle

$$= \frac{|V_s| |V_r|}{|B|} = \frac{(230/\sqrt{3})(230/\sqrt{3})}{85.2}$$

$$= 207 \text{ MVA} = 10.35 \text{ cm}$$

Draw circle diagram (3) with this radius. I is the operating point if no leading MVAR is supplied to SPM. In this case the operating point will be v vertically downward on circle (3)

$\therefore$ Leading MVAR

$$= P_m = 115 \times 20$$

$$= 230 \text{ MVAR per phase}$$

$$= 69 \text{ MVAR for all the 3 diagrams}$$

(c) Maximum power under these conditions can be found by drawing a horizontal line through C and cutting the circle at S' .

Maximum power

$$= SS' = 8.15 \text{ cm}$$

$$8.15 \times 20 = 163.0 \text{ MW for one phase}$$

$$= 489 \text{ MW for all three phases}$$

θ , corresponds to power factor angle θ under this condition and is equal to 50.5° .

$$\therefore \text{Power factor} = \cos 50.5^\circ = 0.636 \text{ (leading)}$$

Example 3.28. A 220 kV, 3 phase, 50 Hz transmission line transmits load of 75 MW at 0.8 p.f. lagging at the receiving end. The A, B, C, D constants are :

$$A = D = 0.9055 \angle 0.6^\circ$$

$$B = 156 \angle 84.5^\circ$$

$$C = 0.00116 \angle 90.2^\circ$$

Determine with the help of universal power circle diagram

- sending end voltage and the phase shift between the sending end and receiving end voltages
- sending end power and line loss
- the maximum power that can be transmitted from the sending end and the maximum power that can be received at the receiving end
- SPM capacity to improve the power factor at the receiving end to 0.85 lagging and
- the improvement in voltage regulation as a result of installation of the SPM referred to in (iv)

Solution. (i) $\angle(\beta - \alpha) = 84.5^\circ - 0.6^\circ = 83.9^\circ$

Let the base value of the voltage be 220 kV

$$nO_R = |A||V_r|^2 = 0.9055 \times \left(\frac{220}{220}\right)^2 = 0.9055 \text{ p.u.}$$

$$\text{Base power} = \frac{|V_r|^2}{|B|} = \frac{(220)^2}{156} = 310 \text{ MVA}$$

$$\therefore \text{Per unit power} = \frac{75}{310} = 0.242$$

Steps to draw the circle diagram (Fig. 3.24)

1. From n draw a line nO_R at an angle of 83.9° from the horizontal, the magnitude of nO_R being 0.9055 p.u.
2. Draw a horizontal line $O_R x$ at the point O_R of magnitude 0.242 p.u.
3. Draw line $O_R P_R$ from O_R at an angle of $\cos^{-1} 0.80$ i.e., 36.9° from the horizontal.
4. Draw a line XP_R perpendicular to $O_R x$ which cuts the line $O_R P_R$ at P_R .

Then nP_R is the radius of the receiving end circle.

From the graph

$$nP_R = 11.35 \text{ cm} = 1.135 \text{ p.u.}$$

$$\text{But } nP_R = |V_s||V_r|$$

$$\therefore V_s = \frac{1.135}{1} = 1.135 \text{ p.u.}$$

$$\therefore \text{Sending end voltage} = 1.135 \times 220 = 250 \text{ kV}$$

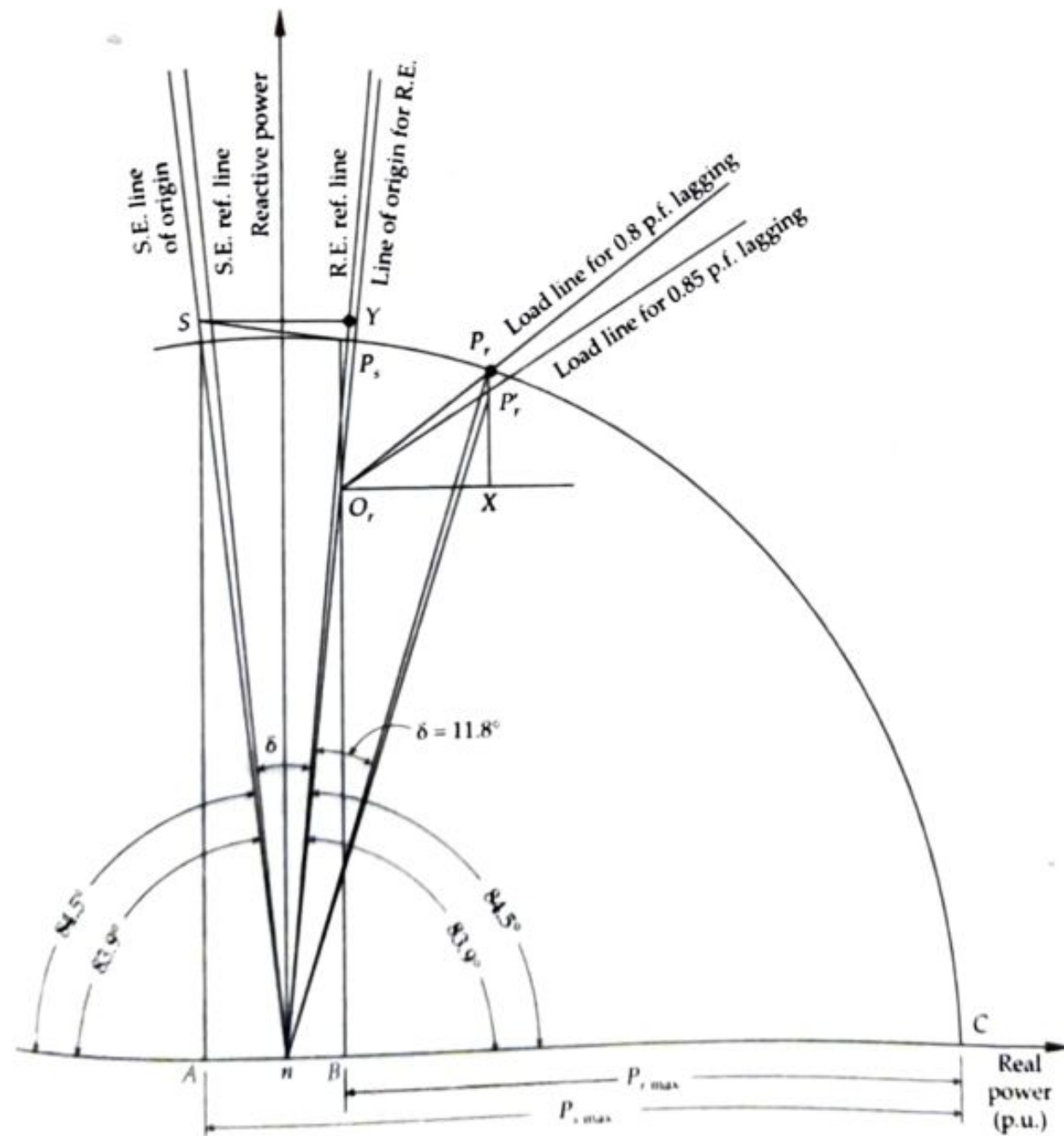
5. Phase shift between sending end and receiving end voltages from the graph = $|P_R nZ| = 11.80^\circ$.

$$\text{Therefore, } \delta = 11.8^\circ$$

(ii) Draw line nO_S from n at angle of 83.9° from the horizontal. As $V_s = 1.135 \text{ p.u.}$

$$\begin{aligned} \text{Distance } n \text{ to } O_s &= |D||V_s|^2 = 0.9055 \times (1.135)^2 \\ &= 1.165 \text{ p.u.} \end{aligned}$$

The sending end reference line and line of origins are drawn on the graph.



The sending end load point P_S must be at a radius of 0.9055 p.u. on the torque line making an angle of 118° with the sending end reference line. From the graph

Sending end real power

$$(O_S Y) = 0.249 \text{ p.u.} = 0.249 \times 310 = 77.2 \text{ MW}$$

Sending end reactive power

$$(Y P_S) = 0.035 \text{ p.u.} \\ = 0.035 \times 310 = 9.92 \text{ MVAR}$$

$$\therefore \text{Line loss} = (77.2 - 75) = 2.2 \text{ MW}$$

(iii) Sending end maximum power (AC)

$$= 1.26 \times 310 = 391 \text{ MW}$$

Receiving end maximum power (BC)

$$1.035 \times 310 = 320.85 \text{ MW}$$

(iv) Draw a load line for 0.85 p.f. lagging. From the graph the SPM capacity to improve the p.f. at the receiving end to 0.85 lagging

$$= P_R P'_R = 0.030 \text{ p.u.} \\ = 0.030 \times 310 = 9.3 \text{ MVAR}$$

$$(v) \quad n P'_R = 1.105 \text{ p.u.}$$

$$\text{But } n P'_R = |V_s| \cdot |V_r|$$

$$\therefore |V_s| = \frac{1.105}{1} = 1.105 \text{ p.u.}$$

i.e., Sending end voltage with the installation of SPM

$$= 1.105 \times 220 = 243.1 \text{ kV}$$

$\therefore$ Voltage regulation

$$= \frac{243.1 - 220}{220} \times 100 = 10.5\%$$

and voltage regulation without the SPM

$$= \frac{250 - 220}{220} \times 100 = 13.6\%$$

In other words the regulation is improved from 13.6 to 10.5% with the installation of the SPM.

Example 3.29. For the transmission line of Example 3.25 draw the loss line if 250 kV is supplied at the sending end and 230 kV is maintained at the receiving end. From the loss line find the line loss when 90 MW is to be supplied to a load at the receiving end. Also find minimum line loss and the receiving end power and power factor at which this occurs.

Solution. As the sending end and receiving end powers are given respectively by equations

$$P_R = \frac{V_s V_r}{B} \cos(\beta - \delta) - \frac{AV_r^2}{B} \cos(\beta - \alpha)$$

$$\text{and } P_S = \frac{DV_s^2}{B} \cos(\beta - \Delta) - \frac{V_s V_r}{B} \cos(\beta + \delta)$$

$\therefore$ Line losses will be the difference of these two powers, i.e.,

$$\begin{aligned} P_L &= P_S - P_R \\ &= \left\{ \frac{DV_s^2}{B} \cos(\beta - \Delta) + \frac{AV_r^2}{B} \cos(\beta - \alpha) \right\} \\ &\quad - \frac{V_s V_r}{B} [\cos(\beta + \delta) + \cos(\beta - \delta)] \\ &= \left\{ \frac{DV_s^2}{B} \cos(\beta - \Delta) + \frac{AV_r^2}{B} \cos(\beta - \alpha) \right\} - 2 \frac{V_s V_r}{B} \cos \beta \cos \delta \\ &= 2 \cos \beta \left[\frac{\frac{DV_s^2}{B} \cos(\beta - \Delta) + \frac{AV_r^2}{B} \cos(\beta - \alpha)}{2 \cos \beta} - \frac{V_s V_r}{B} \cos \delta \right] \end{aligned}$$

Therefore, if a point Q is located at a distance

$$\sigma Q = \frac{DV_s^2 \cos(\beta - \Delta) + \frac{AV_r^2}{B} \cos(\beta - \alpha)}{2 \cos \beta}$$

on the reference line then for any angle δ the projection of point P on the reference line,

$$OR = \frac{V_s V_r}{B} \cos \delta$$

Hence $QR = OQ - OR$ represents to some scale the line losses.

Therefore, if a line NN' is drawn perpendicular to the reference line through Q , the perpendicular from point P to this line, i.e., $PP' = QR$ also represents line losses. The line NN' is thus known as loss line.

For the given data :

$$V_s = 250 \text{ kV}$$

and $V_r = 230 \text{ kV}$

$$A = D = 0.9785 \angle 0.3^\circ$$

$$B = 85.2 \angle 77.47^\circ$$

So that

$$\angle(\beta - \alpha) = 77.47 - 0.3 = 77.17^\circ$$

Choosing the scale of $1 \text{ cm} = 20 \text{ MVA}$, line OO' is drawn equal to 10.11 cm.

$$\text{and } \frac{|A||V_r|^2}{|B|} = \frac{0.9785}{85.2} \times \left(\frac{230}{\sqrt{3}}\right)^2 = 202.25 \text{ MVA}$$

Also,

$$\begin{aligned} \frac{|V_s||V_r|}{|B|} &= \frac{\left(\frac{250}{\sqrt{3}}\right) \times \left(\frac{230}{\sqrt{3}}\right)}{85.2} \\ &= 225 \text{ MVA} = 11.25 \text{ cm} \end{aligned}$$

A circle with centre O is drawn with this radius.

$$OQ = \frac{\frac{|D||V_s|^2}{|B|} \cos(\beta - \Delta) + \frac{|A||V_r|^2}{|B|} \cos(\beta - \alpha)}{2 \cos \beta}$$

$$= \frac{\left[\frac{0.9785}{85.2} \times \left(\frac{250}{\sqrt{3}} \right)^2 \times 0.222 + \frac{0.9785}{85.2} \times \left(\frac{220}{\sqrt{3}} \right)^2 \times 0.222 \right]}{2 \times 0.215}$$

$$= 238 \text{ MVA, i.e., 11.4 cm.}$$

The loss line is drawn perpendicular to the reference line at point Q .

For 90 MW locate point P and from this point length of perpendicular PP' on loss line is incasured 0.2 cm.

$$\therefore \text{Line loss} = (0.2 \times 20) \times 2 \cos \beta$$

$$= 4 \times 2 \times 0.215 = 1.72 \text{ MW per phase}$$

$$= 1.72 \times 3 = 5.16 \text{ MW (total)}$$

As is clear from the diagram the minimum line losses occur at point P_M where reference line cuts the receiving end circle.

From Fig. E3.25,

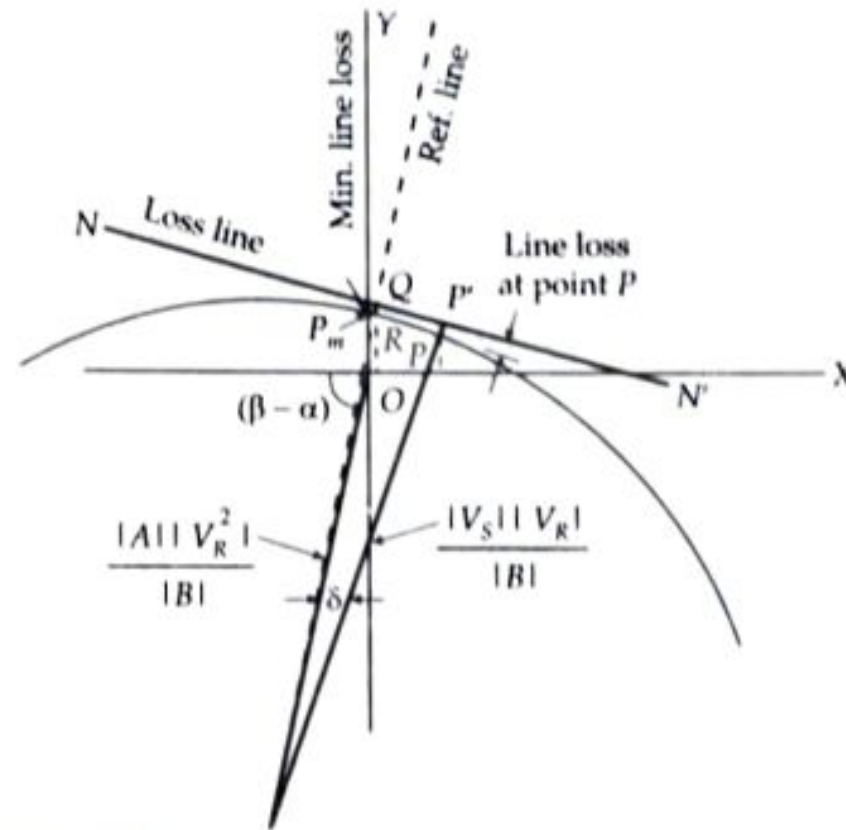


Fig. E3.25

$$\text{Minimum line loss} = (0.13 \times 20) \times 2 \cos \beta$$

$$= 1.118 \text{ MW/phase} = 3.354 \text{ (total)}$$

$$\text{Also, power supplied at this point}$$

$$= 3 \times (0.2 \times 20) = 12 \text{ MW and p.f.} = 0.174$$